EDITION 2026–27

CUET-PG · PAPER CODE SCQP09

Complete Learning-Ordered Syllabus

Computer Science & Information Technology
Subject → Topic → Subtopic → What to learn → How it is asked

225	3	92
ACTUAL QUESTIONS DECODED	PAPERS · 2024, 2025, 2026	STUDY MODULES

A preparation map built from the actual paper — not a generic syllabus. Every module below is tied to real questions, real weightage and the exact way the examiner asks it.

Evidence base: SCQP09 papers of 19 Mar 2024 (Shift 3), 27 Mar 2025 (Shift 2), 11 Mar 2026 (Shift 2), plus the current NTA SCQP09 syllabus. · Prepared 1 November 2026.

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SECTION 1

The exam in numbers

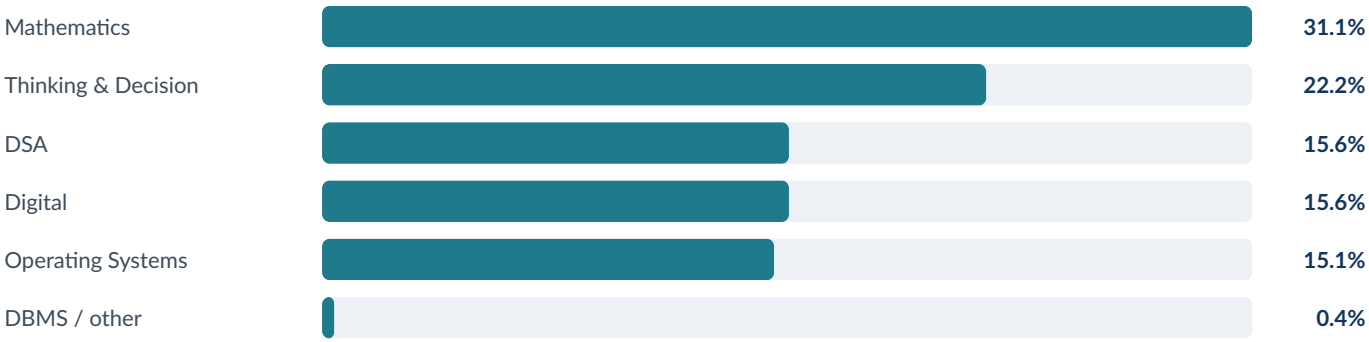
Only three papers matter for this edition: 2024, 2025 and 2026. The paper changed shape in 2024 (all-domain, 75 questions), then changed speed in 2025 (90 minutes). The table below is the factual base for every recommendation in this document.

Paper	Date & shift	Questions	Marks	Duration	Marking	Part A?
2024	19 March 2024 · Shift 3	75	300	105 min	+4 / −1	No
2025	27 March 2025 · Shift 2	75	300	90 min	+4 / −1	No
2026	11 March 2026 · Shift 2	75	300	90 min	+4 / −1	No

Source: first page of each official NTA question paper (duration and total marks are printed on the paper). All questions are compulsory, subject-specific and bilingual.

Subject weightage: the only three-year pattern that matters

Subject block	2024	2025	2026	Total / 225	Average share	What it means
Mathematics	30	20	20	70	31.1%	Largest block; 2024 was an outlier.
Thinking & Decision Making	15	15	20	50	22.2%	Growing; fastest scoring block.
Data Structures & Algorithms	10	13	12	35	15.6%	Stable core.
Digital Fundamentals	10	13	12	35	15.6%	Stable and highly learnable.
Operating Systems	9	14	11	34	15.1%	Volatile but always present.
DBMS / other CS	1	0	0	1	0.4%	One conflict-serializability question.



Five conclusions that drive the whole plan

1. **Mathematics is the largest single block.** 70 of 225 questions. Even in the leaner 2025/2026 papers it is 20 questions — more than any other subject.
2. **TDM is the best return on time.** 50 questions, no prerequisites, and 2026 raised it to 20. It is the block where speed can be trained fastest.
3. **The CS core is remarkably stable.** DSA + Digital + OS = 104 questions (46.2%), and each paper kept the three between 29 and 40 questions.
4. **2026 changed the format, not the syllabus.** Match-the-list, arrange-the-order and assertion–reason questions increased sharply. Content remained the same five blocks.
5. **The clock is now the main enemy.** 90 minutes for 75 questions = 72 seconds per question. The paper rewards recognition and elimination, not long derivations.

What changed across the three papers

Dimension	2024	2025	2026	Preparation implication
Duration	105 min	90 min	90 min	Practise at 72 sec/question, not 84.
Math share	30	20	20	Do not over-index on the 2024 math spike.
TDM share	15	15	20	Treat TDM as a daily subject, not a filler.
OS share	9	14	11	OS numericals are non-negotiable.
Question format	Direct MCQs dominate	More matching	Match + arrange + assertion–reason	Train format conversion, not only content.
General awareness	0	0	4 (anomaly)	Do not start a current-affairs regime.

The three-year DNA, paper by paper

2024 · the math outlier

Math 30 · TDM 15 · DSA 10 · Digital 10 · OS 9

105 minutes. Math alone was 40% of the paper. The CS core was comparatively light, and there was one DBMS question.

Use it for: a hard math drill and a first diagnostic. Do not use its subject split as a forecast.

2025 · the CS-heavy paper

Math 20 · TDM 15 · OS 14 · DSA 13 · Digital 13

90 minutes. The CS core rose to 40 questions; OS had its highest share. Math fell to 20 but remained the largest single subject.

Use it for: OS and DSA numericals, and a 90-minute full mock.

2026 · the format shift

TDM 20 · Math 20 · DSA 12 · Digital 12 · OS 11

90 minutes. TDM became the joint-largest block. Match/arrange/assertion–reason formats increased sharply; four GK questions appeared as an anomaly.

Use it for: format training and TDM speed. It is the closest to the current style.

How to read the three years together. Do not average blindly and do not chase the last paper. The stable core is **Math (20–30) + TDM (15–20) + CS core (30–40)**. Prepare for Math 20 and TDM 20 as the base case, keep enough math depth for a 30-question spike, and assume the CS core will be 30–35. That single planning assumption covers all three papers.

SECTION 2

The learning order

The syllabus is not a list to be studied alphabetically. Some blocks unlock others: algebra unlocks calculus; logarithms unlock complexity; digital memory unlocks OS memory; trees and queues unlock OS scheduling and file systems. The order below follows those dependencies and front-loads the highest-weight, most reusable material.

1. **Mathematics toolkit**
   Sets · counting · probability · statistics · algebra · matrices
2. **Mathematics core**
   Coordinate geometry · calculus (limits, continuity, differentiation, integration)
3. **Digital Fundamentals**
   Number systems · Boolean algebra · combinational · sequential · memory
4. **Data Structures & Algorithms**
   Complexity · linear structures · trees · graphs · sorting · hashing
5. **Operating Systems**
   Processes · scheduling · synchronization · deadlock · memory · virtual memory · I/O
6. **Integrated revision**
   Full papers · error log · formula sheet · speed drills

Parallel track: Thinking & Decision Making does not depend on any other block. Start it on day one and do 30–40 minutes daily alongside the core sequence. Do not wait until the end — it is 22% of the paper and the easiest place to gain speed.

Why this order and not the paper order

Stage	Weeks	What you study	Why it comes here
1. Toolkit	1–3	Sets, counting, probability, statistics, algebra, logs, progressions	Every later math chapter uses these. TDM arithmetic also improves immediately.
2. Math core	4–7	Coordinate geometry, calculus	Calculus is the single largest math topic (21 questions). Coordinate geometry is second (15).
3. Digital	8–9	Number systems, Boolean, K-map, circuits, memory	Short, self-contained, high-scoring. Memory basics are needed for OS.
4. DSA	10–12	Complexity, lists, stacks, trees, graphs, sorting, hashing	Logarithms and summation from math make complexity easy; queues/trees are reused in OS.
5. OS	13–14	Processes, scheduling, deadlock, memory, virtual memory, I/O	Needs DSA (queues, trees) and digital (memory). Best studied after both.
6. Revision	15–16	Full papers, error log, formula sheet, speed	Only works after all five blocks have been seen once.

The six rules behind the sequence

1. Prerequisite first

Do not start calculus before you can solve quadratics, inequalities and functions. Do not start complexity before logarithms and series.

2. Weight before novelty

Math and TDM are 53% of the paper. They get the first eight weeks and daily time, even if OS feels more “interesting”.

3. Short blocks early

Digital is compact and self-contained. Finishing it in two weeks gives an early confidence boost and unlocks OS memory.

4. Numerical before theory

In OS and DSA, learn the formula/algorithm first, then the theory. The paper asks “compute”, “arrange” and “match” more than “define”.

5. TDM daily, not weekly

Reasoning is a skill, not a chapter. Twenty minutes daily beats a four-hour Sunday block.

6. Revise by retrieval

Every module ends with “how it is asked”. Cover the answer, attempt the archetype, then check. Reading notes is not revision.

Stage gates: do not advance until you can…

Stage	Gate test (closed book, timed)
1. Math toolkit	Solve a quadratic, a log equation, an AP/GP question and a three-set Venn problem without notes.
2. Math core	Differentiate and integrate standard forms, find a tangent, and identify a conic from its equation.
3. Digital	Convert bases, fill a K-map, identify a flip-flop, and compute memory capacity from address lines.
4. DSA	State the standard complexities, traverse a tree, and explain BFS vs DFS with their data structures.
5. OS	Compute WT/TAT, check a safe sequence, translate a paging address, and compare page-replacement algorithms.
6. Revision	Attempt 65+ questions at 80%+ accuracy in a strict 90-minute mock, with a written error log.

> **If a gate fails, do not restart the stage.** Repair the specific module that failed, retest it in 48 hours, and keep the rest of the sequence moving. A stage gate is a diagnosis, not a punishment.

SECTION 3

The master syllabus

Each card is one study module. The order inside a subject is the order you should learn it. The three small pills show how many questions came from 2024, 2025 and 2026. The priority badge tells you how much depth to invest.

How to read a card

What to learn is the minimum concept set for that module. If you can explain every bullet and solve one example of each, you are ready for the exam level.

How it is asked is the actual question behaviour observed in the three papers — not a guess.

PYQ signal gives a real question or a close paraphrase so you know the flavour.

Priority legend

P0 5+ questions across three years. Master every bullet. This is where ranks are built.

P1 3–4 questions. Learn the standard patterns and one numerical variant.

P2 1–2 questions. Learn the definition, one formula and one example.

P3 Official but untested recently, or an anomaly. Cover once; do not let it displace P0/P1.

92 modules at a glance

Subject block	Modules	Questions	Share	Largest module (questions)
Mathematics	22	**70**	31.1%	Calculus (21)
Thinking & Decision Making	17	**50**	22.2%	Analogy & classification (13)
Digital Fundamentals	17	**35**	15.6%	Combinational circuits (9)
Data Structures & Algorithms	17	**35**	15.6%	Linear structures (9)
Operating Systems	16	**34**	15.1%	Deadlock (7)
Low-yield official topics	3	**1**	0.4%	DBMS (1)
Total	**92**	**225**	**100%**	Calculus (21)

Counts are the actual 2024–2026 question totals, not estimates. A card is one study module; several cards can share a module name (for example, six calculus cards). “Largest module” aggregates all cards with that module name.

Four rules for using the cards

1. Follow the order. Subject order, then card order, is the learning sequence. Jumping to a later card usually means missing a prerequisite.

2. Read the pills first. 24 / 25 / 26 counts tell you whether a topic is a recurring earner or a one-off. A gray pill is not failure — it is information.

3. Test with “How it is asked”. Close the card and answer each bullet from memory. If you can do that, the module is exam-ready; rereading “What to learn” is not revision.

4. Stop at the depth line. P0 gets every bullet, P1 gets the standard pattern, P2 gets one example, P3 gets one page. Spending P0 time on P3 is how syllabi become unfinished.

3.1 Mathematics

The largest block in all three papers. It is formula-driven and highly predictable: the same eight families reappear with different numbers. The order below goes from the toolkit (sets, counting, algebra) to the two heavyweights (coordinate geometry and calculus).

70	30 / 20 / 20	32 min · 18–22 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. Most questions are one- or two-step. The examiner rarely asks a proof. The difficulty comes from (a) choosing the right formula under time pressure, and (b) match/statement formats that hide four small calculations inside one question. Learn the standard results cold; then drill speed.

M1 Sets, Venn diagrams & inclusion–exclusion

P0 | 24 1 | 25 2 | 26 2

SET THEORY

WHAT TO LEARN

- Set representation, subsets and power set
- Union, intersection, difference and complement
- Cardinality and the inclusion–exclusion principle for 2 and 3 sets
- Venn regions: 'only A', 'A and B but not C', 'neither'
- De Morgan's laws

HOW IT IS ASKED

- Find n(A∪B), n(A∩B) or 'only one game' from survey data
- Three-set Venn diagram with a missing region
- Match List-I (set expression) with List-II (value)
- Word problem: students playing football/cricket, both, neither

PYQ SIGNAL 2024 Q26: 200 students; 120 football, 50 cricket, 30 both — find the number playing exactly one game. 2025 Q17: match n(X), n(Y), n(X∪Y) to n(X∩Y).

M2 Permutations & combinations

P1 | 24 2 | 25 1 | 26 0

COUNTING

WHAT TO LEARN

- Fundamental principle of counting
- nPr and nCr; arrangements with repetition
- Dictionary order / rank of a word
- Circular permutations and gap method
- Selection with constraints: at least, at most, exactly
- Counting lines and triangles from points, including collinear points

HOW IT IS ASKED

- Dictionary rank of a word (e.g., 49th word of AGAIN)
- Number of words with a fixed vowel–consonant pattern
- Number of lines/triangles from n points when some are collinear
- Match formula to value; arrange counting steps

PYQ SIGNAL 2024 Q21: 15 points, 5 collinear — total straight lines. 2024 Q27: 49th word in dictionary order of AGAIN. 2025 Q18: 3 consonants + 3 vowels words.

M3 Basic probability, conditional probability & Bayes

P1 24 1 25 2 26 0

PROBABILITY

WHAT TO LEARN

- Sample space, events, complement
- Addition rule and mutually exclusive events
- Independent vs dependent events
- Conditional probability and multiplication rule
- Total probability theorem and Bayes' theorem
- Coin, dice, card and bag models

HOW IT IS ASKED

- Coin tossed 3 times: P(exactly two heads ∪ at most two tails)
- Bag A/B, one red ball drawn — find the probability it came from a bag (Bayes)
- Statement-based probability identities
- Venn diagram with probability regions

PYQ SIGNAL 2024 Q20: P(A)=1/2, P(B)=7/12, P(not A or not B)=1/4 — which statements are true? 2025 Q34: bag A (3R,4B), bag B (5R,6B), red drawn — Bayes.

M4 Averages, dispersion & frequency distributions

P1 24 1 25 0 26 2

STATISTICS

WHAT TO LEARN

- Mean, median, mode; weighted mean
- Range, mean deviation, variance and standard deviation
- Effect of adding/multiplying every observation
- Combined mean and combined variance
- Frequency distribution, cumulative frequency and ogive
- Coefficient of variation

HOW IT IS ASKED

- Mean and variance of n observations given; find the missing observations
- Correct a wrongly recorded observation and recompute mean/variance
- Match the measure to its value
- Which measure is unaffected by an outlier

PYQ SIGNAL 2024 Q17: mean 5.2, variance 27.296, three values 1,3,6 — find the other two. 2026 Q27: 12 observations, mean 2, variance 4, one value corrected.

M5 Indices, surds & logarithms

P2 24 0 25 1 26 0

ALGEBRA TOOLKIT

WHAT TO LEARN

- Laws of indices and surds
- Definition and laws of logarithms
- Base change
- Logarithmic and exponential equations
- Domain restrictions

HOW IT IS ASKED

- Solve for x in a log equation
- Simplify an expression involving powers
- Match the expression to its value
- Find the number of digits using logarithms

PYQ SIGNAL 2025 Q16: solve a logarithmic equation for x (options 121, 120, 12, 24).

M6 Quadratic equations, inequalities & modulus

P0 24 4 25 1 26 0

ALGEBRA TOOLKIT

WHAT TO LEARN

- Discriminant and nature of roots
- Sum and product of roots; symmetric functions
- Sign of a quadratic and the interval method
- Modulus equations and inequalities
- Integer solutions and counting solutions
- Common roots and formation of equations

HOW IT IS ASKED

- Find the range of a parameter for real roots
- Count positive-integer solutions of a cubic inequality
- Solve abs(x−a)+abs(x−b) ≥ c
- Match equation to root condition

PYQ SIGNAL 2024 Q18: $(x-a)(x-2a)(x-a^2)<0$ has exactly 10 positive-integer solutions — find a. 2024 Q23: roots real ⇒ range of a.

M7 Functions, domain, range & graphs

P2 24 1 25 0 26 0

ALGEBRA TOOLKIT

WHAT TO LEARN

- Domain and range; interval notation
- Composition and inverse functions
- Even/odd, periodic and monotonic functions
- Range of quadratic and trigonometric expressions
- Graph transformations

HOW IT IS ASKED

- Match each function to its range
- Find the domain of a composite/root/log function
- Identify increasing/decreasing intervals
- Find the inverse of a simple function

PYQ SIGNAL 2024 Q36: match functions such as $2-\sin 3x$, (x^2+x+2), $\sin x-\cos x$ to their ranges.

M8 AP, GP, HP & means

P1 24 1 25 1 26 2

ALGEBRA TOOLKIT

WHAT TO LEARN

- AP: nth term, sum, arithmetic mean
- GP: nth term, finite/infinite sum, geometric mean
- HP and harmonic mean
- AM ≥ GM ≥ HM
- Insertion of means and properties of consecutive terms
- Sum of special series

HOW IT IS ASKED

- Given two sequences in GP with the same ratio, what locus do the points lie on?
- If a,b,c are in GP and a^x=b^y=c^z, then x,y,z are in which progression?
- Find a missing term or sum
- Match the progression to its property

PYQ SIGNAL 2024 Q25: $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ with x's and y's in GP — the points lie on a line. 2025 Q22: a,b,c in GP and a^x=b^y=c^z.

M9 Matrix algebra & special matrices

P0 24 1 25 1 26 3

MATRICES

WHAT TO LEARN

- Matrix addition, scalar multiplication and multiplication
- Transpose, symmetric, skew-symmetric, orthogonal
- Idempotent, nilpotent and involutory matrices
- Trace and its properties
- Inverse and transpose identities: $(AB)^T=B^TA^T$, $(AB)^{-1}=B^{-1}A^{-1}$

HOW IT IS ASKED

- Which statements are true/false about symmetric and skew-symmetric matrices?
- Compute $AB-BA$ or A^2 for a given matrix
- Match matrix A to det(A), A^{-1} or A^T
- Find an unknown entry from a matrix equation

PYQ SIGNAL 2024 Q22: if A,B are symmetric, is AB−BA skew-symmetric? 2025 Q21: match matrix expressions to values.

M10 Determinants, inverse & linear systems

P1 24 2 25 1 26 1

MATRICES

WHAT TO LEARN

- Expansion by minors and cofactors
- Properties of determinants; $\det(kA)=k^n\det(A)$; $\det(AB)=\det(A)\det(B)$
- Adjoint and inverse; $A^{-1}=\text{adj}(A)/\det(A)$
- Rank of a matrix and consistency of linear systems
- Cramer's rule; unique/infinite/no solution
- Eigenvalues, trace and determinant

HOW IT IS ASKED

- A determinant identity equals k — find k
- For which k does the system have no/infinite solutions?
- Find $\det(A^{-1})$ or adj(A)
- Use trace/determinant to find eigenvalues

PYQ SIGNAL 2024 Q16: properties of determinants — which statements are true? 2024 Q19: system consistency for parameter k. 2025 Q25: determinant identity equals k.

M11 Straight lines & pair of straight lines

P2 24 1 25 0 26 1

COORDINATE GEOMETRY

WHAT TO LEARN

- Slope forms; point-slope, two-point, intercept form
- Angle between lines; parallel and perpendicular conditions
- Distance from a point to a line
- Family of lines and concurrency
- Pair of straight lines: $ax^2+2hxy+by^2=0$
- Angle bisectors of a pair of lines

HOW IT IS ASKED

- Line through a point with a given sum/product of intercepts
- Find the angle bisectors of a homogeneous pair
- Condition for concurrency
- Area of a triangle formed by three lines

PYQ SIGNAL 2024 Q30: line through (2,3), sum of intercepts 12. 2026 Q25: angle bisectors of $x^2-4xy-5y^2=0$.

M12 Circles

P1 24 1 25 1 26 1

COORDINATE GEOMETRY

WHAT TO LEARN

- Standard and general equation; centre and radius
- Circle through given points
- Tangent, normal, chord and chord of contact
- Family of circles; radical axis
- Length of tangent from an external point

HOW IT IS ASKED

- Find the circle through two points with centre on a given line
- Find centre and radius from $x^2+y^2+2gx+2fy+c=0$
- Find the tangent/normal at a point
- Match the circle to its centre/radius

PYQ SIGNAL 2024 Q33: circle through (3,0),(0,−2), centre on $2x+3y=3$. 2025 Q27: centre and radius of $x^2+y^2+6x-4y+4=0$.

M13 Parabola

P1 24 1 25 1 26 1

COORDINATE GEOMETRY

WHAT TO LEARN

- Standard forms $y^2=4ax$, $x^2=4ay$ and variants
- Focus, directrix, vertex, axis and latus rectum
- Parametric coordinates
- Tangent and normal
- Focal chord properties

HOW IT IS ASKED

- Identify focus/directrix/latus rectum from an equation
- Inscribed triangle or focal chord length
- Find the equation from focus/directrix
- Match the parabola to its properties

PYQ SIGNAL 2024 Q32: equilateral triangle inscribed in $y^2=8x$. 2025 Q30: $x^2=-16y$ — focus, directrix, latus rectum.

M14 Ellipse & hyperbola

P2 24 0 25 1 26 1

COORDINATE GEOMETRY

WHAT TO LEARN

- Standard forms; a,b,c relations
- Major/minor axis, vertices, foci, directrices
- Eccentricity and latus rectum
- Hyperbola asymptotes and rectangular hyperbola
- Conjugate hyperbola
- Foci shared with an ellipse

HOW IT IS ASKED

- Find the major axis and vertices of an ellipse
- Find the hyperbola whose foci coincide with an ellipse's foci
- Compute eccentricity/latus rectum
- Match conic to its eccentricity

PYQ SIGNAL 2025 Q26: major axis and vertices of $3x^2+2y^2=6$. 2026 Q24: hyperbola with foci of a given ellipse.

M15 Conic identification & properties

P2 24 2 25 0 26 0

COORDINATE GEOMETRY

WHAT TO LEARN

- General second-degree equation
- Discriminant B^2-4AC and conic type
- Degenerate conics and pair of lines
- Eccentricity and latus rectum of each conic
- Circle/parabola/ellipse/hyperbola conditions

HOW IT IS ASKED

- Which statements about $ax^2+2hxy+by^2+2gx+2fy+c=0$ are true?
- Match each conic to its eccentricity or latus rectum
- Identify the conic from its equation
- Find the condition for a pair of real distinct lines

PYQ SIGNAL 2024 Q29: conditions for circle vs pair of lines. 2024 Q31: match conics to eccentricity/latus rectum.

M16 Distance, section formula, area & collinearity

P1 24 1 25 2 26 0

COORDINATE GEOMETRY

WHAT TO LEARN

- Distance and section formulae
- Midpoint, centroid and incentre
- Area of a triangle by determinant
- Collinearity and slope condition
- Distance between parallel lines

HOW IT IS ASKED

- Area of a triangle from three vertices
- Find k so that three points are collinear
- Find a point dividing a segment in a ratio
- Match the geometric quantity to its value

PYQ SIGNAL 2024 Q34: area of triangle (1,2),(2,5),(4,3). 2025 Q28: collinearity/ratio problem.

M17 Limits

P0 24 2 25 3 26 1

CALCULUS

WHAT TO LEARN

- Standard limits: $\sin x/x$, $(1-\cos x)/x^2$, $(e^x-1)/x$, $\log(1+x)/x$
- Algebraic manipulation, rationalisation and substitution
- L'Hôpital's rule for 0/0 and ∞/∞
- Series expansions for quick limits
- One-sided limits and limits at infinity
- Indeterminate forms

HOW IT IS ASKED

- Evaluate a 0/0 limit involving roots or trig functions
- Limit at infinity of a rational function
- Match each limit to its value
- Find a parameter so that a limit exists

PYQ SIGNAL 2024 Q39: lim $\sqrt{(1-\cos 2x)}/x$. 2024 Q42: a trigonometric/algebraic limit. 2025: three limit questions.

M18 Continuity & differentiability

P1 24 3 25 1 26 0

CALCULUS

WHAT TO LEARN

- Definition of continuity at a point
- Left-hand and right-hand limits
- Removable, jump and infinite discontinuities
- Finding parameters for piecewise continuity
- Continuity vs differentiability
- Intermediate value theorem (basic)

HOW IT IS ASKED

- Piecewise f(x) is continuous at 0 — find a
- Match f(x) to the value f(0) that makes it continuous
- Which statement about continuity/differentiability is true?
- Find the number of discontinuities

PYQ SIGNAL 2024 Q37: $f(x)=(1-\cos 4x)/x^2$ for $x<0$, a at $x=0$ — find a. 2024 Q40: match functions to f(0).

M19 Differentiation, tangents & normals

P1 24 1 25 0 26 2

CALCULUS

WHAT TO LEARN

- Standard derivatives; product, quotient and chain rules
- Implicit and logarithmic differentiation
- Parametric differentiation: dy/dx=(dy/dt)/(dx/dt)
- Tangent and normal; angle of intersection
- Orthogonal curves
- Rates of change and monotonicity

HOW IT IS ASKED

- Parametric curve — find the tangent's angle with the x-axis
- Two curves cut at right angles — find k
- Find the equation of tangent/normal
- Differentiate an inverse-trig or exponential expression

PYQ SIGNAL 2024 Q35: $x=e^t\cos t$, $y=e^t\sin t$ — tangent angle at $t=\pi/4$. 2026 Q28: $x=y^2$ and $xy=k$ cut orthogonally.

M20 Maxima & minima

P2 24 0 25 0 26 1

CALCULUS

WHAT TO LEARN

- Critical points and stationary points
- First and second derivative tests
- Increasing/decreasing intervals
- Absolute extrema on a closed interval
- Optimisation word problems
- Assertion–reason questions on extrema

HOW IT IS ASKED

- Find the maximum/minimum value of a function
- Identify intervals of increase/decrease
- Assertion–reason about a stationary point
- Optimise area/volume under a constraint

PYQ SIGNAL 2026 Q29: assertion–reason question on maxima/minima.

M21 Integration: substitution, parts & partial fractions

P1 24 2 25 1 26 1

CALCULUS

WHAT TO LEARN

- Standard integrals
- Integration by substitution
- Integration by parts (ILATE)
- Partial fractions
- Trigonometric identities and special substitutions
- Integrals of the form $\int 1/(1+\sqrt{\tan x})$, $\int \sin^2 x$, $\int x\,e^x$

HOW IT IS ASKED

- Evaluate an indefinite integral
- Match each integral to its result
- Choose the correct substitution
- Find $\int f(x)\,dx$ with a given derivative pattern

PYQ SIGNAL 2024 Q43: $\int (x^5-x)^{(1/5)}x^4\,dx$. 2024 Q44: match integrals to results. 2025: integration by parts/partial fractions.

M22 Definite integrals & area

P1 24 2 25 0 26 1

CALCULUS

WHAT TO LEARN

- Fundamental theorem of calculus
- Properties: $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$; even/odd; periodic
- Area between a curve and the x-axis
- Area between two curves
- Intersection points and symmetry
- Definite integral as a limit of a sum (basic)

HOW IT IS ASKED

- Use the a+b−x property to evaluate $\int x\,f(x)$
- Area bounded by $y^2=4x$ and $x^2=4y$
- Evaluate a definite integral using symmetry
- Match the property to its value

PYQ SIGNAL 2024 Q38: if $f(a+b-x)=f(x)$, evaluate $\int x\,f(x)$. 2024 Q45: area between $y^2=4x$ and $x^2=4y$.

3.2 Thinking & Decision Making

Fifty questions, no prerequisites, and the most forgiving block if you train deliberately. It is also the most format-sensitive: series, analogy and coding questions can be solved by pattern recognition in under a minute. Run this as a daily parallel track.

50	15 / 15 / 20	18 min · 16–19 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. The paper tests speed and pattern recognition, not deep logic. Almost every question is a self-contained puzzle. The main risk is spending three minutes on one puzzle; the main opportunity is banking 15+ correct answers in 15 minutes.

T1 Number series

P0 24 3 25 2 26 2

SERIES

WHAT TO LEARN

- Difference and ratio patterns
- Squares, cubes, primes and factorials
- Double differences and alternating series
- Operations on previous terms (×2+1, ×3−2)
- Missing term, next term and nth term
- Wrong-term series

HOW IT IS ASKED

- Find the 10th/nth term of a series
- Find the missing term: 5,17,37,65,?,145
- Two people counting up/down — first common number
- Identify the wrong term

PYQ SIGNAL 2024 Q1: 7,14,28,... 10th term. 2024 Q2: 5,17,37,65,?,145. 2024 Q3: counting-down/up common number.

T2 Letter & alphanumeric series

P1 24 0 25 3 26 1

SERIES

WHAT TO LEARN

- Alphabet positions A=1 ... Z=26 and reverse
- Gaps between letters; increasing gaps
- Letter clusters and alternating series
- Mixed letter–number series
- Next two terms and missing term

HOW IT IS ASKED

- A,C,F,J,?,? — find the next two letters
- D4,F6,H8,J10,? — find the missing term
- Match the letter–number series to its missing term
- Complete the pattern with two interleaved series

PYQ SIGNAL 2025 Q2: A,C,F,J,?,? 2025 Q9: match D4,F6,H8,J10,? and similar series.

T3 Word analogy

P0 24 1 25 3 26 2

ANALOGY & CLASSIFICATION

WHAT TO LEARN

- Relationship types: tool–purpose, cause–effect, part–whole
- Worker–workplace, synonym/antonym, category
- Sound and action relationships
- Direction of the relation must be preserved
- Second-order analogies

HOW IT IS ASKED

- Door : bang :: chain : ? (tinkle/clank)
- Find the missing word in the pair
- Match each pair to its relation
- Choose the pair with the same relation

PYQ SIGNAL 2025 Q3: Door is related to bang as chain is related to tinkle/clank. 2024 Q6: complete the word pair.

T4 Number analogy

P1 24 1 25 0 26 2

ANALOGY & CLASSIFICATION

WHAT TO LEARN

- Same operation on both numbers
- Squares, cubes, digit sums and reversals
- Ratios and differences
- Two-step rules
- Fractions and percentages

HOW IT IS ASKED

- 4:17 :: 7:?
- 720:840 :: 60:?
- 225:256 :: 289:?
- Match each number pair to its missing value

PYQ SIGNAL 2026 Q14: match number analogies such as 4:17::7:?, 720:840::60:?, 225:256::289:?.

T5 Odd one out / classification

P1 24 1 25 1 26 2

ANALOGY & CLASSIFICATION

WHAT TO LEARN

- Semantic categories (months, weapons, occupations)
- Number properties: prime, square, divisibility
- Letter patterns and positions
- Visual/topological properties
- Choose the option that fits the given criterion

HOW IT IS ASKED

- Which word is different? (Arrow, Missile, Sword, Bullet)
- Which month is different? (August, October, November, March)
- Choose the odd figure
- Select the group that is alike

PYQ SIGNAL 2024 Q13: odd month out. 2025 Q13: arrow/missile/sword/bullet — choose the different word.

T6 Coding–decoding

P1 24 1 25 1 26 1

LOGICAL REASONING

WHAT TO LEARN

- Letter shifting (forward/backward)
- Reverse alphabet and mirror codes
- Substitution ciphers
- Number coding and word-to-number mapping
- Conditional coding
- Decode and encode directions

HOW IT IS ASKED

- SCOTLAND=12345678, LOAN=1435 … find the code for a word
- VISHWANATHAN→NAAWTHHSANIV; code KARUNAKARANA
- Find the word for a given code
- Identify the coding rule

PYQ SIGNAL 2024 Q14: multi-word number code. 2025 Q4: VISHWANATHAN letter rearrangement.

T7 Blood relations

P2 24 1 25 1 26 0

LOGICAL REASONING

WHAT TO LEARN

- Family-tree notation
- Generations and gender
- In-law relationships
- 'Only son/daughter', 'brother's father'
- Pointing/Introducing puzzles
- Derive the relation asked, not every relation

HOW IT IS ASKED

- K is the son of A's mother's sister … how is P related to Q?
- A woman introduces a man — how is she related to him?
- Choose the correct relation from the options
- Count the number of males/females

PYQ SIGNAL 2024 Q9: complex family tree. 2025 Q7: 'His brother's father is the only son of my grandfather.'

T8 Direction sense

P2 | 24 1 | 25 0 | 26 0

LOGICAL REASONING

WHAT TO LEARN

- North/East/South/West and turns
- Left/right after a turn
- Net displacement and Pythagoras
- Final direction vs distance
- Shadow and sunrise/sunset direction

HOW IT IS ASKED

- Cycle 15 km south, turn right 7 km, right 10 km, left 5 km — distance to home
- Find the final direction
- Minimum distance between start and end
- Shadow direction at a given time

PYQ SIGNAL 2024 Q11: Amit's cycle route — distance to reach home.

T9 Seating & ranking

P1 | 24 1 | 25 0 | 26 2

LOGICAL REASONING

WHAT TO LEARN

- Linear and circular arrangements
- Left/right and 'between' constraints
- Rank from top/bottom
- Floor and row puzzles
- Elimination and fixed-position method
- Ordering by a measurable attribute

HOW IT IS ASKED

- Four students on a bench — who sits at the corners?
- Arrange games by number of players
- Find the rank from the bottom
- Who is second to the left of X?

PYQ SIGNAL 2024 Q8: bench seating. 2026 Q18: arrange games by team size.

T10 Syllogism

P2 | 24 1 | 25 0 | 26 1

LOGICAL REASONING

WHAT TO LEARN

- All / Some / No statements
- Venn-diagram method
- Conversion and distribution
- Possibility cases
- Conclusions that definitely follow vs may follow
- Complementary pairs

HOW IT IS ASKED

- Statements + conclusions — which follow?
- Two statements + two conclusions
- 'Some pens are books' type chains
- Identify the valid conclusion

PYQ SIGNAL 2024 Q5: pens/books/schools/colleges. 2026 Q4: two statements and two conclusions.

T11 Verbal reasoning & sentence arrangement

P2 | 24 1 | 25 0 | 26 0

LOGICAL REASONING

WHAT TO LEARN

- Para-jumbles and phrase ordering
- Subject–verb–object coherence
- Pronoun reference and connectors
- Statement–assumption/conclusion (basic)
- Cause–effect and course of action

HOW IT IS ASKED

- Arrange four phrases to form a meaningful sentence
- Choose the correct sequence
- Identify the assumption/conclusion
- Find the best conclusion

PYQ SIGNAL 2024 Q10: arrange phrases about Aryabhatta into a correct statement.

T12 Figure series & figure matrix

P1 24 1 25 1 26 2

FIGURE & SPATIAL

WHAT TO LEARN

- Rotation, reflection and movement
- Number of sides/lines and shading
- Addition/subtraction of elements
- 3×3 figure matrix rules
- Two-rule problems (shape + shading)
- Complete the missing cell

HOW IT IS ASKED

- Choose the figure that completes the matrix
- Which figure comes next in the series?
- Select the odd figure
- Identify the rule governing the series

PYQ SIGNAL 2024 Q7: complete the figure matrix. 2025 Q15: next figure in the series.

T13 Figure counting

P2 24 0 25 1 26 0

FIGURE & SPATIAL

WHAT TO LEARN

- Count triangles by size/orientation
- Count squares and rectangles
- Count straight lines and intersections
- Systematic decomposition
- Avoid double counting

HOW IT IS ASKED

- Find the number of triangles in the figure
- Count squares/rectangles
- How many straight lines are needed?
- Choose the correct count from four options

PYQ SIGNAL 2025 Q5: number of triangles (36/44/48/46).

T14 Mirror image, water image & symmetry

P2 24 0 25 0 26 1

FIGURE & SPATIAL

WHAT TO LEARN

- Vertical vs horizontal mirror
- Letters/digits identical to their mirror image
- Water image
- Paper folding and punch holes
- Rotation vs reflection

HOW IT IS ASKED

- Which letters have a mirror image identical to themselves?
- Choose the correct mirror image
- Complete the symmetrical figure
- Paper-folding punch-hole pattern

PYQ SIGNAL 2026 Q16: letters whose mirror image is identical to the letter itself.

T15 Arithmetic puzzles

P2 24 1 25 1 26 0

QUANTITATIVE REASONING

WHAT TO LEARN

- Fractions, percentages and ratios
- Work, time and speed
- Average, age and mixture problems
- Marking/scoring puzzles
- Simple equations from the story
- Unit consistency

HOW IT IS ASKED

- 4/5 boys, 2/5 boys and 1/4 girls absent — what fraction is present?
- 60 questions, +4/−1, score 130 — how many correct?
- Age/work/mixture puzzle
- Find the unknown from a short word problem

PYQ SIGNAL 2024 Q4: absent fraction. 2025 Q11: +4/−1 scoring puzzle.

T16 Missing number in a figure

P2 | 24 1 | 25 1 | 26 0

QUANTITATIVE REASONING

WHAT TO LEARN

- Row/column sums and products
- Squares/cubes and digit operations
- Centre number from surrounding numbers
- Clockwise/anticlockwise patterns
- Two-rule grids

HOW IT IS ASKED

- Find the missing number in the circle/grid
- Choose the missing character
- Identify the operation linking the numbers
- Match the figure to its missing value

PYQ SIGNAL 2024 Q12: missing character in a number grid. 2025 Q6: missing number in a circle (72,24,96 / 16,12,108 / 18,?).

T17 Static science & general knowledge

P3 | 24 0 | 25 0 | 26 4

GENERAL AWARENESS (LOW PRIORITY)

WHAT TO LEARN

- Basic plant/animal cell facts
- Everyday physics and chemistry
- Taxonomy and classification terms
- Famous books, authors, capitals, currencies
- Static GK only; no daily current-affairs regime

HOW IT IS ASKED

- Assertion–reason: leaves are green because of chloroplast/ chromoplast
- Match taxonomy terms (biography, history, soil, nation)
- Odd one out from general knowledge
- Classification of disciplines

PYQ SIGNAL 2026 Q9–Q11: science assertion–reason. 2026 Q13: match taxonomy terms. This is a 2026 anomaly — only 4 questions.

Last-mile TDM drill

Question type	First 30 seconds	Common trap
Number / letter series	Write differences; if not constant, write ratios; then check squares/ cubes and two interleaved series.	Assuming one rule must fit every term.
Analogy	Say the relation as a full sentence, including its direction.	Reversing the pair or using a looser relation.
Odd one out	Find the property shared by three options; the fourth violates it.	Choosing a secondary property instead of the defining one.
Coding–decoding	Write alphabet positions; test forward/backward shift and reversal.	Mixing letter-position and number codes.
Blood relations / direction	Draw the tree or compass immediately; do not hold it in your head.	Confusing left/right after a turn; missing a generation.
Syllogism	Draw the most unfavourable Venn diagram for the conclusion.	Converting “some” into “all”, or assuming existence.

Daily drill: 20 mixed questions in 18 minutes. Log every miss by question type, not by question — the type is what you fix.

3.3 Digital Fundamentals

A compact, rule-based block that rewards accuracy. Number systems, Boolean algebra and K-maps are almost free marks; memory and sequential circuits need a little more care. Study this before OS so that memory terminology is already familiar.

35	10 / 13 / 12	12 min · 9–12 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. The questions are short and formulaic. Most are direct conversions, simplifications, circuit identification or memory facts. There are almost no multi-step numericals. This is the best block for a high-accuracy, low-time pass.

G1 Number systems & base conversion

P2 24 1 25 0 26 0

NUMBER SYSTEMS

WHAT TO LEARN

- Binary, octal, decimal and hexadecimal
- Positional value and conversion
- Radix equations
- BCD, Gray code and Excess-3
- Range of an n-digit number in base r

HOW IT IS ASKED

- $(224)_r = (13)_r$ — find the radix r
- Convert between bases
- Identify the Gray code / BCD equivalent
- Find the largest/smallest number in a given base

PYQ SIGNAL 2024 Q67: $(224)_r = (13)_r$ — find r.

G2 Complements, signed arithmetic & overflow

P2 24 1 25 1 26 0

NUMBER SYSTEMS

WHAT TO LEARN

- 1's and 2's complement
- Signed-magnitude and 2's-complement ranges
- Addition/subtraction using complements
- Overflow detection
- n-bit representation range

HOW IT IS ASKED

- Range of an n-bit 2's-complement number
- Add +6,+13 and −6,+13 in 8-bit 2's complement
- Which result overflows?
- Convert a negative decimal to 2's complement

PYQ SIGNAL 2024 Q69: range of n-bit 2's complement. 2025 Q65: signed-complement addition matching.

G3 Fixed-point & IEEE 754 floating point

P2 24 0 25 2 26 0

NUMBER SYSTEMS

WHAT TO LEARN

- Fixed-point representation
- IEEE 754 single (32-bit) and double (64-bit)
- Sign, biased exponent, mantissa; bias 127/1023
- Normalised/denormalised numbers
- NaN, infinity and comparisons
- Rounding

HOW IT IS ASKED

- What is NaN == NaN?
- Two floats differ only in the sign bit — what is true?
- Identify exponent/mantissa fields
- Find the decimal value of a given IEEE 754 pattern

PYQ SIGNAL 2025 Q70: NaN == NaN. 2025 Q71: two floats differing only in sign bit.

G4 Error detection & Hamming codes

P2 24 0 25 1 26 0

NUMBER SYSTEMS

WHAT TO LEARN

- Parity (even/odd)
- Hamming distance and minimum distance
- Detection vs correction capability
- Hamming code generation and syndrome
- CRC and checksum (basic)

HOW IT IS ASKED

- Minimum Hamming distance to correct up to p errors
- How many parity bits are needed?
- Find the parity bit / codeword
- Detect the error position from the syndrome

PYQ SIGNAL 2025 Q67: minimum Hamming distance to correct p errors (2p+1).

G5 Logic gates & universality

P2 24 0 25 0 26 2

BOOLEAN ALGEBRA

WHAT TO LEARN

- Truth tables of AND, OR, NOT, NAND, NOR, XOR, XNOR
- Universal gates: NAND and NOR
- Gate equivalents and De Morgan
- XOR/XNOR identities
- Gate-level circuit reading

HOW IT IS ASKED

- Which is a universal gate?
- A given circuit generates the same output as which gate?
- Match each gate to its output condition
- Identify the gate from a truth table

PYQ SIGNAL 2026 Q65: universal gate. 2025 Q75: circuit equivalent to XOR/XNOR.

G6 Boolean algebra & simplification

P1 24 1 25 1 26 1

BOOLEAN ALGEBRA

WHAT TO LEARN

- Identity, null, idempotent and complement laws
- Commutative, associative and distributive laws
- Absorption and consensus
- De Morgan's theorems
- Canonical SOP/POS and duality
- Minimise to the fewest variables

HOW IT IS ASKED

- Which Boolean expressions are true?
- Simplify a'b'+ab+a''b to minimum variables
- Find the complement of an expression
- Identify the law used

PYQ SIGNAL 2024 Q68: which Boolean identities are true? 2025 Q73: simplify a'b'+ab+a''b.

G7 Karnaugh-map minimisation

P1 24 1 25 1 26 1

BOOLEAN ALGEBRA

WHAT TO LEARN

- 2-, 3- and 4-variable K-maps
- Gray-code cell ordering
- Grouping 1, 2, 4, 8 cells
- Wrap-around groups
- Don't-care conditions
- Essential prime implicants; SOP and POS

HOW IT IS ASKED

- Find the function represented by the given K-map
- Minimise the K-map expression
- How many essential prime implicants?
- Choose the minimal SOP

PYQ SIGNAL 2024 Q71: K-map → minimal expression. 2025 Q63: 4-variable K-map.

G8 Adders & subtractors

P2 24 1 25 0 26 1

COMBINATIONAL CIRCUITS

WHAT TO LEARN

- Half adder: sum=A⊕B, carry=AB
- Full adder: sum and carry expressions
- Ripple-carry adder
- Half/full subtractor
- Carry-in and overflow
- Parallel adder

HOW IT IS ASKED

- A circuit adds two 1-bit numbers but has no carry input — what is it?
- Full adder inputs/outputs
- Find sum/carry expressions
- Identify the subtractor circuit

PYQ SIGNAL 2024 Q72: half adder definition. 2026 Q64: adder input/output count.

G9 Multiplexers, decoders & encoders

P1 24 0 25 2 26 1

COMBINATIONAL CIRCUITS

WHAT TO LEARN

- 2^n:1 MUX and n select lines
- Implementing Boolean functions with a MUX
- Decoder n:2^n; active-high/low outputs
- Encoder and priority encoder
- Number of 2:1 MUXes to build a larger MUX
- Applications

HOW IT IS ASKED

- Minimum number of 2:1 MUXes for a 16:1 MUX
- Arrange the steps of a 3:8 decoder
- Match the circuit to its function
- Find the output for given select lines

PYQ SIGNAL 2025 Q66: 3:8 decoder steps. 2025 Q72: 2:1 MUXes for 16:1.

G10 Combinational vs sequential logic

P1 24 0 25 2 26 2

COMBINATIONAL CIRCUITS

WHAT TO LEARN

- Combinational: output depends only on present inputs; no memory
- Sequential: output depends on present inputs and past state
- Clock and memory elements
- Identify which circuits are combinational
- Analysis and design flow

HOW IT IS ASKED

- Which of these are combinational circuits? (half adder, full adder vs flip-flop)
- Which are characteristics of a combinational circuit?
- Classify a given circuit
- Compare combinational and sequential

PYQ SIGNAL 2026 Q72–Q73: identify combinational circuits and their properties. 2025 Q68: statements about combinational vs sequential.

G11 Flip-flops & latches

P1 24 1 25 1 26 2

SEQUENTIAL CIRCUITS

WHAT TO LEARN

- SR, JK, D and T flip-flops
- Truth/characteristic tables
- Level vs edge triggering
- Latch vs flip-flop
- Race-around condition
- Excitation tables
- Asynchronous inputs

HOW IT IS ASKED

- Match JK inputs 00/01/10/11 to no-change/set/reset/toggle
- Which flip-flop toggles when J=K=1?
- Identify the triggering type
- Find the next state from a characteristic table

PYQ SIGNAL 2024 Q75: match triggering terms. 2025 Q69: JK flip-flop next-state matching.

G12 Registers & counters

P2 24 0 25 0 26 1

SEQUENTIAL CIRCUITS

WHAT TO LEARN

- Register: store multiple bits
- Shift registers: SISO, SIPO, PISO, PIPO
- Asynchronous/ripple vs synchronous counters
- Modulus of a counter
- Number of flip-flops for 2" states
- Ring and Johnson counters

HOW IT IS ASKED

- Fundamental function of a register
- Number of flip-flops for a mod-N counter
- Match the register type to its data flow
- Find the counter modulus

PYQ SIGNAL 2026 Q68: fundamental function of a register.

G13 Memory hierarchy & cache

P1 24 2 25 1 26 0

MEMORY & ORGANISATION

WHAT TO LEARN

- Registers → cache → main memory → SSD/disk → tape
- Speed, size and cost per bit
- SRAM vs DRAM
- Locality of reference (temporal/spatial)
- Cache hit/miss and hit ratio
- Volatile vs non-volatile

HOW IT IS ASKED

- Arrange memory types by increasing cost per bit
- Cache is based on which principle?
- Which memory is fastest/volatile?
- Match the memory to its property

PYQ SIGNAL 2024 Q70: order DRAM, disk, optical disk, SRAM, tape by cost. 2024 Q73: cache and locality.

G14 RAM, ROM, PLA & memory types

P2 24 2 25 0 26 0

MEMORY & ORGANISATION

WHAT TO LEARN

- RAM: volatile, read/write; SRAM/DRAM
- ROM: mask, PROM, EPROM, EEPROM, flash
- PLA/PAL: programmable combinational logic
- RAM is sequential; PLA is combinational
- Memory cell and chip organisation

HOW IT IS ASKED

- RAM is combinational or sequential? PLA is combinational or sequential?
- Match memory type to property (non-volatile, erasable)
- Which is used for firmware?
- Identify the PLA output

PYQ SIGNAL 2024 Q66: RAM/PLA combinational vs sequential. 2024 Q74: match flash/PMOS/NMOS/CMOS properties.

G15 Memory addressing, MAR/MDR & data lines

P2 24 0 25 1 26 1

MEMORY & ORGANISATION

WHAT TO LEARN

- Capacity = 2^(address lines) × word size
- Address lines vs data lines
- MAR and MDR functions
- Address decoding and chip select
- Read/write sequence
- Word vs byte addressing

HOW IT IS ASKED

- 1 KB memory — how many address/data lines?
- Arrange the steps to retrieve data (MAR → address bus → memory → MDR)
- Match the register to its function
- Find the number of chips needed

PYQ SIGNAL 2025 Q74: address/data lines. 2026 Q75: arrange the memory-read steps.

G16 Addition, subtraction, multiplication & division algorithms

P3 24 0 25 0 26 0

COMPUTER ARITHMETIC (LOW YIELD)

WHAT TO LEARN

- Signed-magnitude and 2's-complement addition/subtraction
- Booth's multiplication
- Restoring and non-restoring division
- Flowcharts and register transfers
- Overflow and shift operations

HOW IT IS ASKED

- Rarely asked directly in 2024–2026
- May appear as a step-ordering or match question
- Identify the algorithm from its flowchart
- Find the number of steps/registers

PYQ SIGNAL No direct question in 2024–2026; keep one page for syllabus completeness.

G17 Integrated circuits & logic families

P3 24 0 25 0 26 0

LOGIC FAMILIES (LOW YIELD)

WHAT TO LEARN

- SSI, MSI, LSI, VLSI scales
- TTL vs CMOS
- Fan-in, fan-out, propagation delay
- Noise margin and power dissipation
- Tri-state logic

HOW IT IS ASKED

- Rarely asked directly
- Match logic family to a property
- Identify the IC scale
- Which family has lowest power?

PYQ SIGNAL No direct question in 2024–2026; cover only if time permits.

Digital: facts that cover most questions

Fact	Value / rule
2's-complement range	n bits: -2^{n-1} to $2^{n-1}-1$. Overflow occurs when the carry into and out of the sign bit differ.
Multiplexer	A 2^n:1 MUX needs n select lines; a tree of 2:1 MUXes for 2^n:1 needs 2^n-1 MUXes.
Decoder	n inputs → 2^n outputs; each output corresponds to one minterm.
Flip-flop states	n flip-flops → 2^n states; a mod-N counter needs $\lceil \log_2 N \rceil$ flip-flops.
Memory capacity	Capacity = 2^(address lines) × word size; address lines = $\log_2$(number of words).
K-map	Groups of 1, 2, 4, 8; Gray-code adjacency; wrap-around allowed; don't-cares may be 0 or 1.
IEEE 754 single	1 sign + 8 exponent (bias 127) + 23 mantissa; normals are $(-1)^s \times 1.M \times 2^{(E-127)}$.

Accuracy rule: label MSB/LSB and the bit width before you convert or add. Most digital errors are position errors, not concept errors.

3.4 Data Structures & Algorithms

The conceptual core of the CS section. The paper favours properties, invariants, complexity and matching over long code traces. Learn the standard data structures, their operations and their complexities, then practise the match/arrange formats.

35	10 / 13 / 12	14 min · 9–12 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. Expect a mix of definitions ("which is not an application of stack"), complexity ordering, tree/graph properties and pseudocode arrangement. Only a few questions require a full trace. Memorise the standard tables and invariants.

D1 Asymptotic notation & order of growth

P1 | 24 1 | 25 1 | 26 2

COMPLEXITY

WHAT TO LEARN

- Big-O, Big-Ω and Big-Θ
- Order of growth; log vs polynomial vs exponential
- Best/average/worst case
- Complexities of sorting, searching and graph algorithms
- Comparing functions; limits and log bases

HOW IT IS ASKED

- Arrange complexities in increasing order
- Match each algorithm to its complexity
- Which is the tightest bound?
- Identify the false complexity statement

PYQ SIGNAL 2024 Q65: order insertion sort best, bubble worst, binary search, merge sort. 2025 Q54: order five operations.

D2 Recurrence relations & recursion

P2 | 24 0 | 25 1 | 26 0

COMPLEXITY

WHAT TO LEARN

- Substitution and recursion-tree methods
- Master theorem
- Common recurrences: T(n)=2T(n/2)+n
- Recursion vs iteration
- Call stack and recursion depth
- Tail recursion

HOW IT IS ASKED

- Match the algorithm to its recurrence
- Solve T(n)=2T(n/2)+O(n)
- Find the complexity of a recursive function
- Arrange recursive calls in order

PYQ SIGNAL 2025 Q53: match binary search/merge sort/quick sort to recurrences.

D3 Arrays, indexing & sparse matrices

P2 | 24 0 | 25 0 | 26 1

LINEAR STRUCTURES

WHAT TO LEARN

- 1D/2D arrays; row-major and column-major order
- Address calculation
- Length from the index set
- Static vs dynamic arrays
- Sparse matrix: triplet representation
- Array applications

HOW IT IS ASKED

- Find the length/number of elements from the index set
- Compute the address of A[i][j]
- How is a sparse matrix stored?
- Which operation is costly in an array?

PYQ SIGNAL 2026 Q63: length of an array from its index set.

D4 Linked lists

P0 | 24 2 | 25 1 | 26 2

LINEAR STRUCTURES

WHAT TO LEARN

- Singly, doubly and circular linked lists
- Node structure and successor/predecessor
- Insertion/deletion at head, tail and middle
- Find the last node; search a key
- Reverse, merge and detect a loop
- Pseudocode and pointer updates

HOW IT IS ASKED

- Which list has every node with a successor?
- Correct code to find the last node
- Arrange pseudocode lines to search key k
- Which pointer changes are needed to delete a node?

PYQ SIGNAL 2024 Q57: successor property. 2024 Q59: find the last node. 2025 Q61: arrange search pseudocode.

D5 Stacks & their applications

P2 | 24 1 | 25 0 | 26 1

LINEAR STRUCTURES

WHAT TO LEARN

- LIFO; push/pop/top
- Array and linked-list implementation
- Infix → postfix/prefix; postfix evaluation
- Balanced parentheses
- Recursion and function calls
- Tower of Hanoi, DFS, backtracking
- Stack permutations

HOW IT IS ASKED

- Which is NOT an application of stack?
- Evaluate a postfix expression
- Check balanced parentheses
- Which permutation can a stack produce?

PYQ SIGNAL 2024 Q61 and 2026 Q52: 'Which is NOT an application of stack?'

D6 Queues, priority queues & heaps

P2 | 24 1 | 25 0 | 26 0

LINEAR STRUCTURES

WHAT TO LEARN

- FIFO; enqueue/dequeue
- Circular queue: full/empty conditions
- Priority queue ADT
- Heap implementation: insert/delete-min
- Applications: CPU scheduling, BFS, Dijkstra
- Deque (basic)

HOW IT IS ASKED

- Which structures implement a priority queue?
- Circular queue full/empty condition
- Queue vs stack in BFS/DFS
- Find the next element removed from a heap

PYQ SIGNAL 2024 Q56: ways to implement a priority queue (array, heap, linked list).

D7 Binary trees, representation & traversals

P2 | 24 0 | 25 0 | 26 1

TREES

WHAT TO LEARN

- Terminology: root, leaf, height, degree
- Full, complete, perfect and skewed trees
- Array and linked representation
- Preorder, inorder, postorder, level order
- Reconstruct a tree from traversals
- Expression trees

HOW IT IS ASKED

- A tree used to represent arithmetic expressions is called?
- Give the preorder/inorder/postorder
- Reconstruct the tree from two traversals
- Find the height/number of leaves

PYQ SIGNAL 2026 Q54: expression tree. Tree traversal/property questions recur across years.

D8 Binary search trees & AVL trees

P0 24 2 25 2 26 1

TREES

WHAT TO LEARN

- BST property; inorder is sorted
- Search, insert and delete
- Worst-case height O(n) for a skewed BST
- AVL balance factor and rotations
- Height O(log n)
- Identify whether a tree is BST/AVL

HOW IT IS ASKED

- Worst-case search in a skewed BST
- Worst-case insert/delete in a BST
- Is the given tree a BST/AVL?
- Which rotation restores balance?

PYQ SIGNAL 2024 Q60: skewed BST search complexity. 2025 Q52: BST insert/delete worst case. 2024 Q62: identify AVL/BST.

D9 Heaps, B-trees & B+ trees

P2 24 1 25 1 26 0

TREES

WHAT TO LEARN

- Min/max heap; complete binary tree
- Array indices: parent i/2, children 2i, 2i+1
- Heapify and heap sort
- B-tree order and node properties
- B+ tree: data at leaves, linked leaves
- Use of B+ trees in databases

HOW IT IS ASKED

- Given index i in a heap, what cannot be computed?
- Best data structure for a database index
- B/B+ tree properties
- Heap sort complexity

PYQ SIGNAL 2024 Q58: B+ tree for a database. 2025 Q62: heap index properties.

D10 Graph representation & basic properties

P1 24 0 25 2 26 1

GRAPHS

WHAT TO LEARN

- Adjacency matrix vs adjacency list
- Directed/undirected; weighted/unweighted
- Degree and handshaking lemma
- Space/time trade-offs
- Sparse vs dense graphs
- Sum of adjacency-list lengths

HOW IT IS ASKED

- For a directed graph, sum of adjacency-list lengths = ?
- Best representation for a sparse graph
- Space required by an adjacency matrix
- Match the algorithm to the data structure used

PYQ SIGNAL 2025 Q51: match BFS/DFS/heap sort/storage to data structures. 2025 Q55: adjacency-list length statements.

D11 BFS, DFS & topological sort

P1 24 0 25 2 26 1

GRAPHS

WHAT TO LEARN

- BFS with a queue; level order
- DFS with a stack/recursion
- Applications: shortest unweighted path, cycle, SCC, bipartite
- Topological sort on a DAG
- Complexity O(V+E)
- BFS vs DFS trade-offs

HOW IT IS ASKED

- Which is NOT an application of DFS?
- BFS always finds the shortest path in an unweighted graph — true?
- Topological order of a DAG
- Which data structure does BFS/DFS use?

PYQ SIGNAL 2025 Q50: not an application of DFS. 2025 Q57: BFS vs DFS shortest path.

D12 Shortest paths, MST & graph algorithm complexity

P2 24 1 25 0 26 0

GRAPHS

WHAT TO LEARN

- Dijkstra; Bellman–Ford; Floyd–Warshall
- Prim and Kruskal MST
- Negative edges and negative cycles
- Complexities with matrix/list/heap
- Topological sort complexity
- When to use which algorithm

HOW IT IS ASKED

- Match algorithm to complexity
- Which algorithm handles negative weights?
- Find the MST weight
- Which algorithm is used for single-source shortest path?

PYQ SIGNAL 2024 Q64: match Bellman–Ford/Dijkstra/Prim/topological sort to complexities.

D13 Sorting algorithms

P2 24 1 25 1 26 0

SORTING & SEARCHING

WHAT TO LEARN

- Bubble, selection, insertion
- Merge sort and quick sort
- Heap sort
- Stability and in-place property
- Best/worst/average complexities
- Number of passes/comparisons
- Pivot selection and randomised quicksort

HOW IT IS ASKED

- Array after k passes of bubble sort
- Quicksort vs randomised quicksort — what differs?
- Which sort is stable/in-place?
- Arrange algorithms by complexity

PYQ SIGNAL 2024 Q63: bubble-sort passes. 2025 Q60: quicksort vs randomised quicksort.

D14 Searching algorithms

P2 24 0 25 0 26 0

SORTING & SEARCHING

WHAT TO LEARN

- Linear search
- Binary search and its precondition (sorted)
- Number of comparisons; mid overflow
- Jump and interpolation search (basic)
- Lower bound for comparison search
- Search in rotated arrays

HOW IT IS ASKED

- How many comparisons does binary search need?
- Which search is independent of n?
- Precondition for binary search
- Find the index of a key

PYQ SIGNAL No standalone question in 2024–2026, but binary search appears inside complexity questions.

D15 Hashing & collision resolution

P2 24 0 25 2 26 0

SORTING & SEARCHING

WHAT TO LEARN

- Hash function, table size, load factor
- Chaining
- Open addressing: linear, quadratic, double hashing
- Clustering
- Universal hashing
- Average O(1) vs worst O(n)

HOW IT IS ASKED

- Search time independent of n — which technique?
- Elements hashing to the same slot are stored in a linked list — which method?
- Which collision-resolution method suffers clustering?
- Find the hash index

PYQ SIGNAL 2025 Q56: search time independent of n. 2025 Q59: chaining.

D16 Cross-topic DSA matching

P2 24 0 25 0 26 2

INTEGRATED DSA

WHAT TO LEARN

- Map algorithm ↔ data structure ↔ application
- BFS/queue, DFS/stack, Dijkstra/priority queue
- MST/Prim/Kruskal
- Sparse graph/adjacency list
- Topological sort/DAG
- Recursion/stack

HOW IT IS ASKED

- Match List-I (MST, stack, BFS, sparse graph) with List-II
- Cross-topic one-to-one matching
- Which combination is correct?
- Identify the odd pairing

PYQ SIGNAL 2026 Q57–Q58: cross-topic matching of algorithms, data structures and applications.

D17 Functions, recursion & parameter passing

P3 24 0 25 0 26 0

PROGRAMMING BASICS (LOW YIELD)

WHAT TO LEARN

- Function call and activation record
- Pass by value vs pass by reference
- Recursion and the call stack
- Scope and storage classes (basic)
- Array as a parameter
- C-style pseudocode

HOW IT IS ASKED

- Output of a recursive function
- Number of function calls
- Pass-by-value vs pass-by-reference output
- Arrange pseudocode lines

PYQ SIGNAL No direct 2024–2026 question; official syllabus includes it. Cover with recursion basics.

DSA: the complexity table to write from memory

Operation / algorithm	Best	Average	Worst
Binary search (sorted array)	O(1)	O(log n)	O(log n)
Merge sort	O(n log n)	O(n log n)	O(n log n)
Quick sort	O(n log n)	O(n log n)	$O(n^2)$
Heap sort	O(n log n)	O(n log n)	O(n log n)
BST search	O(1)	O(log n)	O(n)
Hash search	O(1)	O(1)	O(n)
BFS / DFS	—	O(V+E)	O(V+E)
Dijkstra (binary heap)	—	O((V+E) log V)	O((V+E) log V)

Memory hook: "log n for halving, n log n for sorting by comparison, n^2 for pairwise work, V+E for graph traversal." If an option violates the hook, it is usually wrong.

3.5 Operating Systems

The most numerical CS block: scheduling, paging, EAT, Banker's algorithm and page replacement. It is also the most volatile in weight (9–14), so do not skip it in a lean year. The theory questions are usually matching or classification.

34	9 / 14 / 11	14 min · 8–11 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. Half the block is formula-based, half is terminology. The numericals are repetitive once you know the formula; the theory questions test whether you can distinguish similar concepts (deadlock vs starvation, paging vs segmentation, prevention vs avoidance).

O1 OS functions, types & DOS/UNIX/Windows

P2 24 0 25 1 26 0

FUNDAMENTALS

WHAT TO LEARN

- OS as resource manager and interface
- Kernel, shell, system programs
- Batch, multiprogramming, time-sharing, real-time
- Monolithic, microkernel, layered
- DOS/UNIX/Windows architecture
- Boot process

HOW IT IS ASKED

- Arrange MS-DOS layers from innermost to outermost
- Which OS type is this?
- Kernel vs shell
- Match the OS feature to the system

PYQ SIGNAL 2025 Q49: arrange MS-DOS layers (ROM BIOS, device drivers, resident system program, MS-DOS).

O2 Processes, threads & fork()

P2 24 1 25 0 26 1

FUNDAMENTALS

WHAT TO LEARN

- Process states and PCB
- Context switching
- Threads: ULT vs KLT; multithreading models
- fork() return values; number of children
- Zombie, orphan and daemon
- Process vs thread

HOW IT IS ASKED

- fork() called n times — how many child processes?
- Which thread termination model is described?
- Process vs thread differences
- Match the state to its transition

PYQ SIGNAL 2024 Q52: fork() called 4 times — number of children. 2026 Q42: target thread termination.

O3 System calls, interrupts & modes

P2 24 0 25 0 26 1

FUNDAMENTALS

WHAT TO LEARN

- User mode vs kernel mode
- System call mechanism and trap
- Interrupt vs exception
- Types of system calls
- Mode switch and context save
- Timer interrupt

HOW IT IS ASKED

- An event signaled by a software special instruction is a ___
- Which is a system call?
- Interrupt vs trap
- What happens on a system call?

PYQ SIGNAL 2026 Q41: event signaled by a special software instruction (trap).

O4 Inter-process communication (IPC)

P2 24 0 25 0 26 1

FUNDAMENTALS

WHAT TO LEARN

- Shared memory vs message passing
- Pipes and named pipes
- Sockets and signals
- Producer–consumer with IPC
- Synchronization requirements
- Advantages/trade-offs

HOW IT IS ASKED

- Match shared memory/message passing to its description
- Which IPC is fastest?
- Pipe vs socket
- Which model needs synchronization?

PYQ SIGNAL 2026 Q43: match shared memory/message passing to fundamental IPC models.

O5 CPU scheduling algorithms

P1 24 0 25 3 26 1

CPU SCHEDULING

WHAT TO LEARN

- FCFS, SJF, SRTF, priority, round robin
- Preemptive vs non-preemptive
- Criteria: WT, TAT, RT, throughput, utilisation
- Gantt charts
- Convoy effect, starvation, aging
- Short/medium/long-term scheduler
- Multilevel queue/feedback

HOW IT IS ASKED

- Which algorithm prefers the shortest burst?
- Match algorithm to feature
- Compute average waiting/turnaround time
- Which scheduler is the CPU scheduler?

PYQ SIGNAL 2025 Q36: which scheduler is also called CPU scheduler? 2025 Q40: shortest-burst algorithm. 2025 Q46: match algorithms to features.

O6 Critical section, semaphores & monitors

P1 24 1 25 1 26 1

SYNCHRONIZATION

WHAT TO LEARN

- Critical-section problem
- Mutual exclusion, progress, bounded waiting
- Peterson's algorithm
- Test-and-set, compare-and-swap
- Semaphores: counting/binary; wait/signal
- Mutex, monitor, condition variable
- Busy waiting

HOW IT IS ASKED

- Match critical region/circular wait/working set/condition variable to concepts
- Arrange critical-section code
- Semaphore value after operations
- Which condition is violated?

PYQ SIGNAL 2024 Q53: match terms. 2025 Q48: arrange critical-section code.

O7 Classical synchronization problems

P2 24 0 25 1 26 1

SYNCHRONIZATION

WHAT TO LEARN

- Producer–consumer (bounded buffer)
- Readers–writers
- Dining philosophers
- Semaphore solution structure
- Order of wait/signal
- Deadlock and starvation in these problems

HOW IT IS ASKED

- Dining philosophers can be solved by?
- Arrange producer semaphore operations
- Which problem uses a bounded buffer?
- Find the error in the given solution

PYQ SIGNAL 2025 Q41: dining philosophers. 2026 Q49: arrange producer semaphore operations.

O8 Deadlock conditions, prevention & avoidance

P0 | 24 1 | 25 2 | 26 2

DEADLOCK

WHAT TO LEARN

- Coffman conditions: mutual exclusion, hold-and-wait, no preemption, circular wait
- Resource-allocation graph
- Prevention by breaking a condition
- Avoidance and safe state
- Deadlock vs starvation
- Minimum resources to avoid deadlock

HOW IT IS ASKED

- 4 processes each need 5 units — minimum units to avoid deadlock
- A situation where processes wait for each other is called?
- Match the condition to the strategy
- Which is not a Coffman condition?

PYQ SIGNAL 2024 Q48: minimum resource units. 2025 Q37: definition of deadlock.

O9 Banker's algorithm & safe sequence

P2 | 24 1 | 25 0 | 26 0

DEADLOCK

WHAT TO LEARN

- Allocation, Max, Available, Need
- Need = Max − Allocation
- Safety algorithm
- Safe sequence and safe state
- Resource-request algorithm
- Why Banker's is avoidance, not detection

HOW IT IS ASKED

- Given allocation/max/available, which statements are true?
- Find a safe sequence
- Can the request be granted?
- Compute the Need matrix

PYQ SIGNAL 2024 Q49: allocation/max/available table — evaluate statements.

O10 Deadlock detection & recovery

P2 | 24 0 | 25 1 | 26 0

DEADLOCK

WHAT TO LEARN

- Wait-for graph and cycle detection
- Detection algorithm frequency
- Recovery: process termination
- Resource preemption, rollback, victim selection
- Starvation during recovery
- Detection vs prevention vs avoidance

HOW IT IS ASKED

- Which is a way to recover from a deadlock that already occurred?
- How is a deadlock detected?
- Choose the victim
- Which graph is used?

PYQ SIGNAL 2025 Q43: recovery from an occurred deadlock (process termination).

O11 Contiguous allocation & fragmentation

P2 | 24 0 | 25 1 | 26 0

MEMORY MANAGEMENT

WHAT TO LEARN

- Fixed vs variable partitioning
- First-fit, best-fit, worst-fit
- Internal vs external fragmentation
- Compaction
- Swapping and relocation
- Memory protection

HOW IT IS ASKED

- External fragmentation occurs when?
- Which allocation policy leaves the smallest hole?
- Internal vs external fragmentation
- What does compaction do?

PYQ SIGNAL 2025 Q38: when does external fragmentation occur?

O12 Paging & address translation

P1 | 24 1 | 25 1 | 26 1

MEMORY MANAGEMENT

WHAT TO LEARN

- Pages and frames; page table
- Logical → physical address
- Page offset bits = $\log_2$(page size)
- Virtual address bits = page-number bits + offset
- TLB and EMAT
- Multilevel and inverted page tables
- PTBR

HOW IT IS ASKED

- How many bits for the virtual address?
- Find page number/frame number/offset
- Translate a logical address
- TLB hit ratio and EMAT

PYQ SIGNAL 2024 Q47: 1K pages, 512 frames, 2 KB page — virtual address bits. 2025 Q44: 32 KB logical, 4 KB page, 64 KB physical.

O13 Segmentation

P2 | 24 1 | 25 0 | 26 0

MEMORY MANAGEMENT

WHAT TO LEARN

- Segment table: limit and base
- Logical address = segment number + offset
- Bounds check: offset < limit
- Segmentation vs paging
- Sharing and protection
- External fragmentation

HOW IT IS ASKED

- Given a segment table, find the physical address
- Which check must pass?
- Segmentation vs paging
- What fault occurs for an invalid offset?

PYQ SIGNAL 2024 Q51: segment table — physical address for a logical address.

O14 Demand paging, EAT & thrashing

P1 | 24 2 | 25 0 | 26 1

VIRTUAL MEMORY

WHAT TO LEARN

- Demand paging and page faults
- Effective access time: EAT=(1−p)m+p·s
- Page-fault rate and service time
- Locality and working set
- Thrashing and its causes
- Belady's anomaly (FIFO)

HOW IT IS ASKED

- Page fault service 10 ms, memory 30 ns, 1 fault per 10^6 — find EAT
- Decreasing RAM leads to?
- What causes thrashing?
- Compute the page-fault rate

PYQ SIGNAL 2024 Q46: EAT with 10 ms service, 30 ns access, 10^{-6} fault rate. 2024 Q55: decreasing RAM → more page faults.

O15 Page replacement algorithms

P2 | 24 1 | 25 1 | 26 0

VIRTUAL MEMORY

WHAT TO LEARN

- FIFO, LRU, Optimal
- Reference string and frame count
- Belady's anomaly
- Stack algorithms
- Counting page faults and hits
- LRU implementation

HOW IT IS ASKED

- Count page faults for a reference string with 3 frames
- Does increasing frames always reduce faults?
- Which algorithm is optimal?
- Which suffers Belady's anomaly?

PYQ SIGNAL 2025 Q45: reference string 7,0,1,2,0,3,0,4,2,3,0,3,2 with 3 frames. 2024 Q54: frame count vs FIFO faults.

O16 File systems, disk scheduling & I/O

P1 24 0 25 2 26 1

FILE & I/O

WHAT TO LEARN

- File allocation: contiguous, linked, indexed
- Directory and inode
- Free-space management
- Disk scheduling: FCFS, SSTF, SCAN, C-SCAN, LOOK
- Seek, rotational and transfer time
- I/O registers, polling, interrupts, DMA
- Device drivers

HOW IT IS ASKED

- Which disk algorithm chooses the closest track?
- Match the performance metric to its definition
- Match device registers to descriptions
- Arrange the I/O steps

PYQ SIGNAL 2025 Q39: closest-track algorithm (SSTF). 2025 Q47: match seek/transfer metrics. 2026 Q50: device registers.

OS: the five numerical templates

Numerical	Template / check
CPU scheduling	TAT = CT − AT; WT = TAT − BT; draw the Gantt chart before computing. For RR, keep the ready queue in arrival order.
Deadlock bound	n processes, each needing at most m units: n(m−1)+1 units guarantee no deadlock.
Paging bits	Offset = $\log_2$(page size); page-number bits = $\log_2$(number of pages); virtual bits = page bits + offset.
EAT	EAT = (1−p)·m + p·s. Convert ns/ms first; the answer is usually dominated by p·s.
Page replacement	Simulate the reference string frame by frame; count faults, not hits. FIFO can show Belady's anomaly; LRU cannot.

One-line check: every OS numerical has a “before you calculate” condition — preemptive or not, offset < limit, Need = Max − Allocation. Write it down first.

OS theory traps

Pair	Distinction the examiner tests
Deadlock vs starvation	Deadlock: a set of processes can never proceed. Starvation: a process can proceed in principle but is indefinitely overlooked.
Paging vs segmentation	Paging is fixed-size, invisible to the programmer, internal fragmentation. Segmentation is variable-size, logical, external fragmentation.
Prevention vs avoidance	Prevention breaks one Coffman condition. Avoidance uses future information (Banker's) to stay in a safe state.
Detection vs recovery	Detection finds the cycle (wait-for graph). Recovery breaks it: terminate a process or preempt a resource.
Internal vs external fragmentation	Internal: allocated block is larger than needed. External: free memory is split into non-contiguous holes.
Page fault vs thrashing	A page fault is one miss. Thrashing is a sustained high fault rate that leaves no CPU time for useful work.

3.6 Low-yield official topics

These topics appear in the official syllabus but barely appear in the three papers. They are included so the syllabus is complete, not because they deserve equal time. Cover them once, late, with a one-page summary each.

1	1 / 0 / 0	3 min · 0–1 correct
QUESTIONS · 2024–26	YEAR-WISE SPLIT	TARGET TIME / SCORE

Nature of the block. Do not build a full DBMS or CN preparation block from this document. The evidence is clear: 1 DBMS question in 225, 0 CN questions. If your target year's official syllabus explicitly includes them, keep a layer/protocol table and a transaction/normalization summary; otherwise, skip after one reading.

Read this before spending time here. The current NTA syllabus (December 2025) lists Computer Networks and DBMS. The 2024–2026 papers tested one DBMS question and zero CN questions. Treat these as insurance, not as a main subject. The opportunity cost of a full DBMS/CN block is enormous: those hours are better spent on calculus, TDM or OS numericals.

R1 DBMS: transactions & conflict serializability

P3 | 24 1 | 25 0 | 26 0

DBMS

WHAT TO LEARN

- ACID properties
- Schedule and conflicting operations
- Conflict serializability
- Precedence graph and cycle test
- Serial vs non-serial schedule
- Basic 2PL (context)

HOW IT IS ASKED

- A schedule is conflict serializable if …
- Draw the precedence graph
- Which schedule is conflict serializable?
- Identify conflicting operations

PYQ SIGNAL 2024 Q50: definition of conflict serializability. Only DBMS question in 225 recent questions.

R2 DBMS: ER model, normalization, SQL & relational algebra

P3 | 24 0 | 25 0 | 26 0

DBMS

WHAT TO LEARN

- ER diagrams and cardinality
- Relational model, keys and constraints
- Functional dependencies; 1NF–3NF/BCNF
- Relational algebra and calculus
- SQL basics; Codd's rules
- Transactions/recovery (overview)

HOW IT IS ASKED

- Rarely/never asked in 2024–2026
- Could appear as a definition or match question
- Normalize a relation
- Identify the key/constraint

PYQ SIGNAL No question in 2024–2026 beyond Q50's serializability. Keep a 2-page orientation, not a full DBMS block.

R3 Computer networks (official but untested recently)

P3 24 0 25 0 26 0

COMPUTER NETWORKS

WHAT TO LEARN

- OSI and TCP/IP layers
- LAN/Ethernet/token ring; transmission media
- IP(v4) addressing and routing
- TCP/UDP; flow and error control
- DNS, HTTP, SMTP, POP, FTP
- Switches, routers, firewalls; basic cryptography

HOW IT IS ASKED

- 0 questions in 2024–2026
- If it returns: layer matching, protocol-to-layer, IP/subnet, device matching
- Keep a one-page layer/protocol table

PYQ SIGNAL No CN question in the 225 recent questions. Official Dec-2025 syllabus lists CN; do not skip the layer table entirely.

The only CN table you need

TCP/IP layer	Protocols / devices	One-line job
Application	HTTP, DNS, SMTP, POP3, FTP, DHCP	User services, naming and mail/web transfer.
Transport	TCP, UDP	End-to-end delivery, ports, reliability (TCP) or speed (UDP).
Internet	IP, ICMP, routing protocols	Logical addressing, packet forwarding and error reporting.
Network access	Ethernet, Wi-Fi, ARP, hub, switch, bridge	Frames on the local link; MAC addressing.

Device line: hub broadcasts, switch forwards by MAC, router forwards by IP, gateway translates between protocols. OSI adds session and presentation layers; TCP/IP folds both into application. If CN appears, expect layer/protocol matching — not subnetting.

DBMS insurance in six lines

- **ACID:** Atomicity (all or nothing), Consistency (constraints preserved), Isolation (concurrent schedules behave serially), Durability (committed data survives failure).
- **Conflict serializability:** build a precedence graph; the schedule is conflict-serializable iff the graph is acyclic.
- **Keys:** superkey ⊇ candidate key; primary key is the chosen candidate; foreign key references a candidate key.
- **Normal forms:** 1NF atomic values; 2NF no partial dependency on a composite key; 3NF no transitive dependency; BCNF every determinant is a candidate key.
- **Relational algebra:** σ select rows, π project columns, ⋈ join, × product, ∪/∩/− set operations.
- **SQL order:** FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY. Know one aggregate query with GROUP BY and HAVING.

Stop line: this page is the whole DBMS insurance policy. Do not open query optimisation, recovery algorithms or normalization proofs unless the official syllabus forces them.

SECTION 4

How questions are asked

Content knowledge is only half the paper. The other half is recognising the format quickly. Across 225 questions, ten recurring formats account for almost everything. The percentages below are keyword-based approximations; they are useful for training, not for predicting an exact count.

Format	Approx. count	Share	What it looks like	Time target
Direct single-concept MCQ	~120	~53%	One definition, one formula or one short numerical; four options.	45–60 s
Match List-I with List-II	28	12.4%	Four pairs; one-to-one matching; often 4 small facts in one question.	75–90 s
Arrange in correct order	21	9.3%	Algorithm steps, OS layers, semaphore operations, memory read sequence.	60–75 s
Multi-statement true/false	14	6.2%	Statements A, B, C, D; choose the correct combination.	75–90 s
Assertion–Reason	9	4.0%	A and R; decide truth and whether R explains A. Concentrated in 2026.	60 s
“Which is NOT” / odd one out	~15	~7%	Three correct examples and one exception; or an odd item from a group.	45–60 s
Diagram / figure based	~15	~7%	Figure series, counting, K-map, tree/graph diagram, circuit.	60–90 s
Cross-topic matching	~10	~4%	Algorithm ↔ data structure ↔ application; concept ↔ example.	75–90 s

Formats overlap: a question can be both “match” and “multi-statement”. Counts come from keyword tagging of the 225 classified questions, then manual review of ambiguous cases.

The ten archetypes, with real examples

A1 · Direct definition / recall

45–60 s · ~40% of paper

“Which of the following is a universal logic gate?” (2026 Q65) · “Which is NOT an application of stack?” (2024 Q61)

Approach: Answer from a memorised table, not from first principles. If two options look similar, the examiner is testing a distinction (NAND vs NOR, stack vs queue, RAM vs ROM). **Trap:** “NOT” questions — mark the word before reading the options.

A2 · One-formula numerical

60–90 s · ~25%

“Page-fault service time 10 ms, memory access 30 ns, one fault per 10^6 accesses — find EAT.” (2024 Q46) · “1K pages, 512 frames, 2 KB page — how many virtual address bits?” (2024 Q47)

Approach: Write the formula, substitute in consistent units, estimate the answer before computing. **Trap:** ns vs ms; bits vs bytes; page size vs number of pages.

A3 · Match List-I with List-II

75–90 s · 28 questions

“Match JK flip-flop inputs 00/01/10/11 with no-change/set/reset/toggle.” (2025 Q69) · “Match algorithms with recurrences.” (2025 Q53)

Approach: Do the pairs you are sure of first; eliminate options; use the one-to-one constraint. **Trap:** One wrong pair can make the whole option wrong; do not guess the last pair blindly.

A4 · Arrange in correct order

60–75 s · 21 questions

"Arrange the semaphore operations for the producer." (2026 Q49) · "Arrange the steps to retrieve data from memory." (2026 Q75)

Approach: Find the fixed first and last step, then order the middle. Use the dependency, not the option order. **Trap:** Two steps can look interchangeable; check what must be true before the next.

A5 · Multi-statement true/false

75–90 s · 14 questions

"RAM is a combinational circuit; PLA is a combinational circuit…" (2024 Q66) · "Which statements about symmetric matrices are true?" (2024 Q22)

Approach: Evaluate each statement independently and mark T/F; then match the pattern to the options. **Trap:** One false word ("always", "only", "every") invalidates a statement.

A6 · Assertion–Reason

60 s · 9 questions, all 2026

"A: Leaves are green. R: Plants contain chromoplast." (2026 Q9)

Approach: First decide A and R separately; then ask whether R is the correct explanation of A. **Trap:** A true R that explains a different fact is not the answer. In 2026 this format appeared in science GK and calculus.

A7 · "Which is NOT" / odd one out

45–60 s · ~15 questions

"Which is NOT an application of stack?" (2024 Q61, 2026 Q52) · "Which word is different: arrow, missile, sword, bullet?" (2025 Q13)

Approach: Test each option against the defining property; the exception usually violates one criterion. **Trap:** Several options may share a secondary property; use the most essential one.

A8 · Diagram / figure based

60–90 s · ~15 questions

"Find the number of triangles." (2025 Q5) · "Complete the figure matrix." (2024 Q7) · "Find the function from the K-map." (2024 Q71)

Approach: Label the figure, count systematically by size/orientation, and look for one rule at a time. **Trap:** Double counting; assuming rotation when the rule is reflection.

A9 · Multi-statement combination

75–90 s · common in math

"Choose the correct answer from the options: (1) A and B only (2) B and C only …"

Approach: This is A5 with a different wrapper. Evaluate the statements, then eliminate combinations. **Trap:** Options often repeat one statement; use it as an anchor.

A10 · Cross-topic matching

75–90 s · 2026 increase

"Match: MST, stack, BFS, sparse graph with height-balanced tree, recursion, shortest paths, adjacency list." (2026 Q57)

Approach: Build a three-column mental table (concept → structure → application). Start with the pair you know best. **Trap:** The same term can fit two descriptions; use the unique one.

Question wording decoder

Phrase in the paper	What it usually means	First action
“Which of the following statements is/are true?”	Multi-statement T/F; one false word can flip the answer.	Mark each statement T/F before looking at options.
“Choose the correct answer from the options given below”	Combination options (A/B/C/D only).	Eliminate by one confident statement.
“Arrange … in the correct order/sequence”	Process or algorithm steps.	Find first/last, then dependencies.
“Match List-I with List-II”	Four one-to-one pairs.	Fix sure pairs; use elimination.
“Which is NOT / except”	Three true, one false.	Underline NOT; test the defining property.
“Assertion (A) … Reason (R) …”	Truth of A, truth of R, and explanation link.	Decide A, then R, then the link.
“The minimum number of … required”	Extremal formula (MUX, resources, bits, states).	Write the known formula; check units.
“In the light of the above statements”	Assertion–reason or data-sufficiency style.	Do not import outside assumptions.

Top traps by subject

Subject	Most common trap	How to avoid it
Mathematics	Using a formula outside its condition (L'Hôpital without 0/0; area without absolute value; conic orientation).	Write the condition next to the formula; check the sign/orientation before computing.
TDM	Reversing the relation in analogy; missing “only/at least”; choosing a secondary odd-one-out criterion.	Say the relation in a full sentence before applying it; underline quantifiers.
Digital	MSB/LSB order; bias 127 vs 1023; grouping K-map cells across the edge; combinational vs sequential.	Label bit positions; draw the K-map with Gray order; ask “does it have memory?”
DSA	Best vs worst case; 0-index vs 1-index; directed vs undirected edge counts; complete vs full tree.	Write the case and the convention at the top of the rough work.
OS	Preemptive vs non-preemptive; Need = Max − Allocation; offset ≥ limit; FIFO Belady anomaly.	Write the algorithm’s precondition; recheck units and table columns.

SECTION 5

Weightage and trends

The subject split tells you what to study; the topic split tells you what to master. The table below aggregates the 225 questions by module. Read it as a priority list, not a fixed prediction: the paper rotates individual subtopics but keeps the same families.

The 20 highest-yield modules

Subject	Module	2024	2025	2026	Total
Mathematics	**Calculus**	10	5	6	**21**
Mathematics	**Coordinate geometry**	6	5	4	**15**
Thinking & Decision Making	**Analogy & classification**	3	4	6	**13**
Thinking & Decision Making	**Logical reasoning**	6	2	4	**12**
Mathematics	**Algebra toolkit**	6	3	2	**11**
Thinking & Decision Making	**Series**	3	5	3	**11**
Mathematics	**Matrices**	3	2	4	**9**
Digital Fundamentals	**Combinational circuits**	1	4	4	**9**
Data Structures & Algorithms	**Linear structures**	4	1	4	**9**
Digital Fundamentals	**Boolean algebra**	2	2	4	**8**
Data Structures & Algorithms	**Trees**	3	3	2	**8**
Digital Fundamentals	**Memory & organisation**	4	2	1	**7**
Data Structures & Algorithms	**Graphs**	1	4	2	**7**
Operating Systems	**Deadlock**	2	3	2	**7**
Thinking & Decision Making	**Figure & spatial**	1	2	3	**6**
Digital Fundamentals	**Number systems**	2	4	0	**6**
Mathematics	**Set Theory**	1	2	2	**5**
Digital Fundamentals	**Sequential circuits**	1	1	3	**5**
Data Structures & Algorithms	**Complexity**	1	2	2	**5**
Operating Systems	**Fundamentals**	1	1	3	**5**

Modules are groups of closely related subtopics (for example, all calculus subtopics are one module). The full subtopic split is in the master syllabus cards.

What is stable, what is volatile

Stable (plan for these every year)

- Calculus: 5–10 questions every year.
- Coordinate geometry: 4–6 every year.
- Series, analogy and logical reasoning: 10–14 together.
- Digital Boolean/K-map/circuits: 6–9.
- OS deadlock + memory + virtual memory: 6–9.
- DSA trees + graphs + complexity: 6–9.

Volatile (do not over-fit to one year)

- Mathematics: 30 in 2024, then 20 and 20. Do not assume 30.
- OS: 9 → 14 → 11. A lean year can still ask 14.
- TDM: 15 → 15 → 20. The trend is upward, not fixed.
- General awareness: 0 → 0 → 4. Treat 2026 as an anomaly.
- DBMS/CN: 1 → 0 → 0. Insurance only.

Subtopic heat map

Band	Subtopics	What to do
Hot (5–7 Qs)	Number series, word analogy, limits, matrix algebra, sets/Venn, deadlock	Master every variant; these alone are ~15% of the paper.
Warm (3–4 Qs)	Asymptotic order, BST, flip-flops, progressions, quadratic, continuity, integration, odd-one-out, figure series, letter series, linked lists, graphs, Boolean, K-map, MUX, memory hierarchy, CPU scheduling, paging, synchronization, virtual memory, circles, parabola, determinants, statistics	Learn the standard pattern and one numerical variant each.
Cool (1–2 Qs)	The long tail: hashing, heaps, B-trees, sorting, probability, P&C, ellipse, conic properties, file systems, segmentation, banker's algorithm, etc.	Know the definition and one example; do not chase every variant.
Cold (0 Qs)	CN, deep DBMS, computer arithmetic algorithms, IC logic families, parameter passing	One-page summary late; no full block.

The 80/20 of this paper

If you master only the **top 20 modules** in the table above, you cover **179 of 225 questions (79.6%)**. Add the next 10 modules and you are above 90%. The remaining long tail is real, but it is not where ranks are decided.

SECTION 6

High-yield vs low-yield

A syllabus is also a list of things not to study. This section is deliberately blunt: it separates what the paper rewards from what the syllabus mentions but the paper ignores. Use it to protect your time.

Tier 1 — master these first (P0)

Mathematics

- ☐ Limits and continuity (10 Qs)
- ☐ Differentiation, tangents and maxima/minima (4 Qs)
- ☐ Integration and definite integrals/area (7 Qs)
- ☐ Coordinate geometry: circle, parabola, ellipse/hyperbola, lines (15 Qs)
- ☐ Matrices and determinants (9 Qs)
- ☐ Sets/Venn and quadratic/inequalities (10 Qs)

TDM

- ☐ Number and letter series (11 Qs)
- ☐ Word/number analogy and odd one out (13 Qs)
- ☐ Coding–decoding, blood relations, direction, seating (9 Qs)
- ☐ Syllogism and verbal reasoning (3 Qs)
- ☐ Figure series, counting and mirror images (6 Qs)

Digital

- ☐ Number systems, complements, floating point (6 Qs)
- ☐ Boolean algebra, gates and K-map (8 Qs)
- ☐ Combinational circuits: adders, MUX, decoder (8 Qs)
- ☐ Flip-flops, registers, counters (6 Qs)
- ☐ Memory hierarchy, RAM/ROM, addressing (7 Qs)

DSA + OS

- ☐ Complexity and recurrences (5 Qs)
- ☐ Trees: BST, AVL, B/B+, heaps (8 Qs)
- ☐ Graphs: representation, BFS/DFS, shortest paths (7 Qs)
- ☐ Linked lists, stacks, queues (8 Qs)
- ☐ Sorting, searching, hashing (6 Qs)
- ☐ Deadlock and Banker's algorithm (7 Qs)
- ☐ Memory, paging, virtual memory (10 Qs)
- ☐ CPU scheduling, synchronization, file/disk (12 Qs)

Tier 2 — cover once, late (P2/P3)

Topic	Why it is Tier 2	Minimum coverage
Hashing, heaps, B/B+ trees	2 Qs each; often definitional.	One page of properties + one complexity table.
Probability, P&C, statistics	3 Qs each; formula-driven but lower frequency.	Standard formulas + one solved example each.
Ellipse, hyperbola, conic identification	2 Qs each; overlap with coordinate geometry.	Standard forms, eccentricity, latus rectum.
File systems, disk scheduling, segmentation	1–3 Qs; often one numerical.	Disk algorithms + segment translation + inode basics.
Banker's algorithm	1 Q in 2024; still a classic numerical.	Need matrix + safety check; one solved table.
Computer arithmetic, logic families, DBMS transactions	0–1 Qs each; official but absent or rare.	One-page flowchart/property summary; conflict serializability + precedence graph.

What to drop without guilt

- **English grammar, vocabulary, reading comprehension.** These belonged to the pre-2024 Part A. There is no Part A in 2024–2026.
- **Daily current affairs and general studies.** 2026 added four static-science/GK questions; that is an anomaly, not a new section. A 15-minute weekly static-GK review is enough.
- **Deep DBMS:** normalization proofs, SQL query tuning, recovery algorithms, Codd's rules in detail. One question in three years.
- **Computer networks:** subnetting drills, routing protocol internals, cryptography mathematics. Zero questions in three years. Keep only a layer/protocol table as insurance.
- **Software engineering, TOC, compilers, AI/ML, web development, Python/Java.** Not in the current SCQP09 syllabus and not in the papers.
- **Advanced engineering mathematics:** differential equations, complex analysis, 3D geometry, vector calculus, linear programming, real-analysis proofs. Not asked.
- **Advanced computer architecture:** pipelining hazards, cache-mapping numericals, microprogramming, instruction formats. Only memory-hierarchy basics appear.
- **Hardware electronics:** transistor-level analysis, IC fabrication, logic-family electrical characteristics. Only the digital-logic abstraction is asked.

Official but untested: the honest position

The December 2025 NTA syllabus lists **Computer Networks** and a full **DBMS** block. The 2024–2026 papers do not reflect that breadth: 0 CN and 1 DBMS question. If your target year's syllabus still lists them, do a **two-hour orientation** (layers/protocols; ER/normalization/transactions) in the final month. Do not let it become a third CS subject. The evidence says the marks are in the five core blocks.

If CN/DBMS are in your syllabus: the two-hour insurance plan

Block	60-minute minimum	What to skip
Computer Networks	OSI and TCP/IP layer table; protocol-to-layer matching (HTTP, DNS, SMTP, FTP, TCP, UDP, IP); LAN devices (hub, switch, bridge, router, gateway); transmission media; IPv4 basics; firewall/cryptography definitions.	Subnetting drills, routing-algorithm calculations, congestion-control maths, protocol-header bit fields.
DBMS	ACID; conflict serializability and precedence graph; ER entities/relationships; keys and constraints; 1NF–3NF/BCNF definitions; relational algebra symbols; one SQL SELECT example.	Query optimisation, recovery algorithms, security/authorisation, object-relational details, Codd's rules in depth.

When to do it: in the last 10 days, not in the first 10 weeks. Spend 60 minutes on each block, make a one-page sheet, and test yourself with 10 definitional questions. If the paper again ignores these topics, you have lost two hours — not two months.

SECTION 7

The 16-week plan

This plan assumes 4–5 focused hours on weekdays and 6–7 on weekends. If you have less time, use the 6-week crash plan below. The sequence is fixed; the pace is adjustable.

Week-by-week

Week	Core block (2.5–3 h/day)	Parallel TDM (30–40 min/day)	Weekly test
1	Math toolkit: sets, Venn, counting, probability	TDM: series + analogy (daily)	Diagnostic: 2024 paper, untimed
2	Algebra: indices/logs, quadratic, inequalities, functions, AP/GP/HP	TDM: coding, blood relations, direction	30 math + 20 TDM questions
3	Statistics + matrices/determinants	TDM: seating, syllogism, verbal	Sectional: math toolkit
4	Coordinate geometry: lines, circle	TDM: figure series + counting	Sectional: coordinate
5	Coordinate geometry: parabola, ellipse, hyperbola, conics	TDM: mixed figures + puzzles	Full math section, timed
6	Calculus: limits, continuity	TDM: 20-question speed set	Sectional: calculus I
7	Calculus: differentiation, maxima/minima, integration, area	TDM: error log review	Full math paper, 35 min
8	Math revision + weak-topic repair	TDM: full 20 in 18 min	Half mock: math + TDM
9	Digital: number systems, complements, floating point, error codes	TDM daily	Sectional: digital I
10	Digital: Boolean, K-map, combinational, sequential, memory	TDM daily	Sectional: digital II
11	DSA: complexity, arrays, linked lists, stacks, queues	TDM daily	Sectional: DSA I
12	DSA: trees, BST, AVL, heaps, B-trees	TDM daily	Sectional: DSA II
13	DSA: graphs, sorting, searching, hashing	TDM daily	Sectional: DSA III
14	OS: processes, scheduling, synchronization, deadlock	TDM daily	Sectional: OS I
15	OS: memory, paging, virtual memory, file/disk + full revision	TDM daily	Full mock #1
16	Full revision: formula sheet, error log, 4 full mocks	TDM: 20 in 15 min	Full mocks #2–#4 + analysis

Plan rules that matter more than the schedule

- **Never skip two TDM days in a row.** Reasoning is a skill; a four-day gap costs more than a missed chapter.
- **Every week ends with a timed test.** Untimed practice measures knowledge; timed practice measures rank.
- **If a week slips, cut practice volume — not the concept block.** Finishing the concept once is cheaper than restarting it later.
- **Protect the last 20 minutes of the day.** Retrieval and the error log are where marks actually move.

The daily template

Weekday (4–5 h)

- **60 min** — new concept + 10 solved examples.
- **60 min** — 25–30 practice questions on that concept.
- **40 min** — TDM sprint (20 questions, timed).
- **40 min** — previous-day retrieval + error log.
- **30 min** — formula sheet / flashcards.

Weekend (6–7 h)

- **2 h** — sectional test (75 questions or one subject).
- **2 h** — full analysis: every wrong answer, every guess.
- **2 h** — weak-topic repair from the error log.
- **1 h** — TDM mixed set + formula revision.

6-week crash plan

Week	What to do	Non-negotiable
1	Math toolkit + limits/continuity + TDM series/analogy	30 math + 20 TDM questions daily
2	Calculus (differentiation/integration/area) + coordinate geometry	One full math section, timed
3	Matrices, sets, probability, statistics + Digital number systems/Boolean	Digital K-map and conversion drills
4	Digital circuits/memory + DSA complexity/lists/trees	DSA complexity table memorised
5	DSA graphs/sorting/hashing + OS scheduling/deadlock	Two OS numericals daily
6	OS memory/virtual memory + 4 full mocks + error log	One full mock every alternate day

Revision rhythm: 1–3–7–21

> Revise each new module on **day 1** (same evening), **day 3** (short retrieval), **day 7** (mixed questions) and **day 21** (full-paper context). This is more effective than rereading notes. Use the priority badge: P0 modules get all four touches; P2/P3 get only day 1 and day 21.

Weekly scoreboard

Metric	Target by week 8	Target by week 16
Questions attempted in 90 min	65	72–75
Accuracy on attempted	75%	85%+
TDM time for 20 questions	22 min	15–17 min
Math time for 20 questions	35 min	28–30 min
Full mock score	180–200	230–250+

SECTION 8

Exam-day strategy

A 90-minute, 75-question paper with +4/−1 is a triage exercise. You cannot solve everything; you must choose what to solve, in what order, and when to leave. The plan below is built for the actual subject mix.

The three-pass method

Pass	Time	What to attempt	Rule
Pass 1	30–35 min	TDM, digital, direct DSA/OS facts, easy math.	If it takes more than 60 seconds, mark and move. Bank the sure shots.
Pass 2	35–40 min	One/two-step numericals, match/arrange, statement questions.	Use elimination; write only the essential rough work.
Pass 3	15–20 min	Flagged questions, educated guesses, review of NOT/units.	Never leave a question you have eliminated two options on.

Recommended attempt order

If you are strong in math

1. TDM (15–20 Qs, 15 min) — warm-up and confidence.
2. Digital (10–13 Qs, 11 min) — high accuracy.
3. Math (20–30 Qs, 30 min) — the big block while fresh.
4. DSA (10–13 Qs, 13 min).
5. OS (9–14 Qs, 12 min).
6. Review (9 min).

If math is your weak area

1. TDM (15–20 Qs, 16 min).
2. Digital (10–13 Qs, 11 min).
3. DSA (10–13 Qs, 13 min).
4. OS (9–14 Qs, 13 min).
5. Math (20–30 Qs, 28 min) — attempt the easy 15 first.
6. Review (9 min).

The guessing rule (do the maths once)

With four options and +4/−1, a blind guess has expected value **0**. If you eliminate one option, EV = **+0.33**; eliminate two, EV = **+1.0**; eliminate three, mark it. Therefore:

- **No elimination → leave it.** A zero-EV guess adds variance, not marks.
- **One option eliminated → guess only if you are short on attempts.**
- **Two options eliminated → always guess.**
- **Three options eliminated → never leave it.**

Ten instructions for the hall

1. **Read the last line first.** It tells you whether the question is “which is true”, “which is NOT”, “minimum”, or “arrange”.
2. **Write the subject and formula before solving.** Ten seconds of labelling prevents most silly errors.
3. **Use units as a check.** ns vs ms, bits vs bytes, degrees vs radians, sq. units.
4. **Do not re-derive standard results.** Recall the limit, the derivative, the complexity, the flip-flop table.

5. **For match questions, fix the sure pairs first.** The one-to-one constraint usually finishes the rest.
6. **For arrange questions, find first and last.** The middle often follows.
7. **For statement questions, mark T/F in the margin.** Do not hold four evaluations in your head.
8. **For numericals, estimate before computing.** If the options are far apart, approximation is enough.
9. **Never spend more than 2 minutes on one question in Pass 1.** The paper is designed to make you do that.
10. **Keep the last 5 minutes for review.** Check NOT questions, marking, and that no easy question is blank.

What a top score looks like

75 questions, 300 marks. A realistic AIR-1 range is **240–270**: about 62–68 correct, 3–5 wrong and 4–8 blank. That is not perfection; it is **triage plus accuracy**. The plan above is designed to get you to 65 attempts with 90% accuracy, not to make you solve every question.

The final 10-day checklist

Day	Focus	Output
10	Math mixed set: calculus, coordinate, matrices, algebra	30 questions + rewrite the formula sheet from memory
9	Digital: number systems, K-map, circuits, memory	25 questions + one-page error list
8	DSA: complexity, trees, graphs, sorting, hashing	25 questions + complexity table
7	OS: scheduling, deadlock, paging, EAT, page replacement	25 questions + all formulas on one page
6	TDM sprint: series, analogy, coding, figures	40 questions in 35 minutes
5	Full mock #1, strict 90 minutes	Score + question-level analysis (2 hours)
4	Repair the three weakest topics from the mock	20 questions per weak topic
3	Full mock #2, strict 90 minutes	Score + update error log
2	Formula sheet, top traps, 20 TDM questions	No new topics; only retrieval
1	Light revision, exam logistics, sleep	No mock; no new material; 7–8 hours sleep

On exam morning: eat a normal breakfast, carry the admit card and ID, reach the centre early, and do a 5-minute warm-up of easy TDM questions — not a new math chapter.

APPENDIX A

Formula and fact sheet

The minimum set to revise in the last 48 hours. If a formula is not here, it was not worth memorising for this paper.

Mathematics

Concept	Formula / fact
Quadratic	Roots of $ax^2+bx+c=0$: $x = [-b \pm \sqrt{(b^2-4ac)}] / 2a$; sum = $-b/a$; product = c/a; real roots $\Leftrightarrow b^2-4ac \geq 0$.
AP / GP / HP	AP: $a_n=a+(n-1)d$, $S_n=n/2[2a+(n-1)d]$. GP: $a_n=ar^{n-1}$, $S_n=a(1-r^n)/(1-r)$, $S\infty=a/(1-r)$ for $\lvert r\rvert<1$. HM: reciprocals are in AP.
Counting	nPr = $n!/(n-r)!$; nCr = $n!/[r!(n-r)!]$; $nC_0=nCn=1$; $nCr=nC(n-r)$.
Probability	$P(A\cup B)=P(A)+P(B)-P(A\cap B)$; $P(A\lvert B)=P(A\cap B)/P(B)$; Bayes: $P(B_i\lvert A)=P(B_i)P(A\lvert B_i)/\Sigma P(B_j)P(A\lvert B_j)$.
Statistics	Mean = $\Sigma x/n$; variance = $\Sigma(x-\bar{x})^2/n$; SD = $\sqrt{\text{variance}}$; combined mean = $(n_1\bar{x}_1+n_2\bar{x}_2)/(n_1+n_2)$.
Lines	Distance = $\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$; area = $\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$; point-line distance = $\lvert ax_0+by_0+c\rvert/\sqrt{(a^2+b^2)}$.
Circle	$x^2+y^2+2gx+2fy+c=0$: centre $(-g,-f)$, radius $\sqrt{(g^2+f^2-c)}$.
Parabola	$y^2=4ax$: focus $(a,0)$, directrix $x=-a$, latus rectum $4a$. $x^2=4ay$: focus $(0,a)$, directrix $y=-a$.
Ellipse	$x^2/a^2+y^2/b^2=1$ $(a>b)$: $c^2=a^2-b^2$, $e=c/a$, latus rectum $2b^2/a$; major axis $2a$.
Hyperbola	$x^2/a^2-y^2/b^2=1$: $c^2=a^2+b^2$, $e=c/a$, asymptotes $y=\pm(b/a)x$; latus rectum $2b^2/a$.
Limits	$\lim(\sin x/x)=1$; $\lim((1-\cos x)/x^2)=\frac{1}{2}$; $\lim((e^x-1)/x)=1$; $\lim(\log(1+x)/x)=1$; L'Hôpital only for 0/0 or ∞/∞.
Derivatives	$(\sin x)'=\cos x$; $(\cos x)'=-\sin x$; $(\tan x)'=\sec^2 x$; $(e^x)'=e^x$; $(\ln x)'=1/x$; $(x^n)'=nx^{n-1}$.
Tangents	Slope at a point = dy/dx; normal slope = $-1/(dy/dx)$; parametric $dy/dx=(dy/dt)/(dx/dt)$.
Integrals	$\int x^n dx=x^{n+1}/(n+1)+C$; $\int e^x dx=e^x+C$; $\int \sin x\,dx=-\cos x+C$; $\int \cos x\,dx=\sin x+C$; $\int 1/x\,dx=\ln\lvert x\rvert+C$.
Definite integrals	$\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$; $\int_{-a}^{a} f(x)dx=0$ if f is odd, $2\int_0^a f(x)dx$ if f is even.
Matrices	$\lvert kA\rvert=k^n\lvert A\rvert$; $\lvert AB\rvert=\lvert A\rvert\lvert B\rvert$; $A^{-1}=\text{adj}(A)/\lvert A\rvert$; $(AB)^T=B^TA^T$; $(AB)^{-1}=B^{-1}A^{-1}$; trace = Σ eigenvalues; $\lvert A\rvert$ = Π eigenvalues.

Digital Fundamentals

Concept	Formula / fact
2's complement range	n bits: -2^{n-1} to $2^{n-1}-1$.
Number of states	n flip-flops → 2^n states; mod-N counter needs $\lceil \log_2 N \rceil$ flip-flops.
Hamming distance	Detect d errors: dmin ≥ d+1. Correct p errors: dmin ≥ 2p+1.

Concept	Formula / fact
Multiplexer	2^n:1 MUX has n select lines; a 2:1 MUX tree for 2^n:1 needs 2^n-1 MUXes.
Memory capacity	Capacity = 2^(address lines) × word size; address lines = $\log_2$(number of words).
K-map	Groups of 1, 2, 4, 8; Gray-code adjacency; wrap-around allowed; don't-cares can be 0 or 1.
Flip-flops	JK: 00 no change, 01 reset, 10 set, 11 toggle. D: next state = D. T: toggle when T=1.
IEEE 754 single	1 sign + 8 exponent (bias 127) + 23 mantissa; value = $(-1)^s \times 1.M \times 2^{(E-127)}$ for normals.

Operating Systems

Concept	Formula / fact
Scheduling	TAT = CT − AT; WT = TAT − BT; response time = first CPU − AT.
Deadlock bound	n processes, each needs at most m units: minimum units to guarantee no deadlock = n(m−1)+1.
Banker's	Need = Max − Allocation; a state is safe if a sequence exists that finishes all processes with Available.
Paging bits	Offset bits = $\log_2$(page size); page-number bits = $\log_2$(number of pages); virtual bits = page bits + offset; frame bits = $\log_2$(number of frames).
EAT (page fault)	EAT = (1−p)·m + p·s, where p = page-fault rate, m = memory access time, s = page-fault service time.
EAT (TLB)	EAT = h(t+m) + (1−h)(t+2m), where h = TLB hit ratio, t = TLB lookup time, m = memory access.
Segmentation	Physical address = base + offset, provided offset < limit; otherwise trap.
Disk	Access time = seek + rotational latency + transfer; rotational latency = ½ rotation for average.

Data Structures & Algorithms

Concept	Formula / fact
Tree invariant	Any non-empty binary tree: $n_0 = n_2 + 1$ (leaves = degree-2 nodes + 1).
Full binary tree	Height h (root at 0): max nodes = 2^(h+1)−1; max leaves = 2^h.
Heap indices	1-indexed: parent = $\lfloor i/2 \rfloor$, left = 2i, right = 2i+1. 0-indexed: parent = $\lfloor (i-1)/2 \rfloor$, left = 2i+1, right = 2i+2.
Graph counts	Directed adjacency list total length = |E|; undirected = 2|E|; adjacency matrix space = $O(V^2)$.
Master theorem	T(n)=aT(n/b)+f(n); compare f(n) with n^(log_b a): Case 1/2/3.
Sorting	Bubble/selection/insertion $O(n^2)$; merge O(n log n); quick avg O(n log n), worst $O(n^2)$; heap O(n log n).

Concept	Formula / fact
Graph algorithms	BFS/DFS O(V+E); Dijkstra O((V+E)log V) with heap; Bellman–Ford O(VE); Floyd–Warshall O(V^3); Prim O(E log V); Kruskal O(E log E).
Searching	Linear O(n); binary O(log n), requires sorted input; hashing average O(1), worst O(n).

Thinking & Decision Making: quick rules

Type	Rule
Number series	Check differences → ratios → squares/cubes → alternating → operations on previous terms. Verify the rule on every term.
Letter series	Write A=1 … Z=26; check gaps; check reverse alphabet; check two interleaved series.
Analogy	Say the relation as a sentence (“X is used for Y”); apply the same direction to the second pair.
Syllogism	Draw Venn diagrams; “some A are B” does not give “all A are B”; test the conclusion in the most unfavourable diagram.
Blood relations	Draw a tree; mark gender; use “only son of my grandfather” to fix the father.
Direction	Fix N/E/S/W; right turn = clockwise; use Pythagoras for net displacement.
Seating	Start with the most constrained position; use elimination; for circular, fix one person and orient left/right.
Figure series	Check rotation, reflection, number of elements, shading, and position separately.

Numbers worth memorising

Set	Values
Powers of 2	2^0=1, 2^1=2, 2^2=4, 2^3=8, 2^4=16, 2^5=32, 2^6=64, 2^7=128, 2^8=256, 2^9=512, 2^{10}=1024, 2^{11}=2048, 2^{12}=4096, 2^{16}=65,536, $2^{20}\approx10^6$, $2^{30}\approx10^9$.
Factorials	0!=1, 1!=1, 2!=2, 3!=6, 4!=24, 5!=120, 6!=720, 7!=5040, 8!=40,320.
Squares (11–30)	121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900.
Cubes (1–12)	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728.
Logarithms	$\log_{10}2$=0.3010, $\log_{10}3$=0.4771, $\log_{10}5$=0.6990, $\log_{10}7$=0.8451; ln2=0.6931, ln10=2.3026.
Primes up to 50	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.
2's-complement ranges	4-bit: −8 to +7; 8-bit: −128 to +127; 16-bit: −32,768 to +32,767; 32-bit: −2,147,483,648 to +2,147,483,647.
Memory units	1 KB=2^{10} B; 1 MB=2^{20} B; 1 GB=2^{30} B; 1 ms=10^6 ns; 1 µs=10^3 ns.

These are the numbers that turn a 90-second calculation into a 20-second recognition. Drill them as flashcards, not as a list to read.

APPENDIX B

Method and sources

This document is evidence-based, not copied from a coaching syllabus. Here is exactly what was used and how the numbers were produced.

Primary evidence

Paper	File	Pages	Questions	Role
2024, 19 March, Shift 3	SCQP09_2024_March19_Shift3.pdf	85	75	Full classification
2025, 27 March, Shift 2	SCQP09_2025_March27_Shift2.pdf	65	75	Full classification
2026, 11 March, Shift 2	SCQP09_2026_March11_Shift2.pdf	64	75	Full classification

All three are official NTA papers. The 2022 and 2023 papers were excluded because the pattern changed (100 questions with a Part A general section, then 75 all-domain questions from 2024) and the evidence base for this edition is deliberately 2024–2026.

Official syllabus

The current NTA SCQP09 syllabus (December 2025) was used for the official-scope check: Thinking & Decision Making, Mathematics, Operating Systems, Data Structures, Digital Fundamentals — plus Computer Networks and DBMS. The 2024–2026 papers measure which of these are actually tested. Source: NTA CUET-PG syllabus, Computer Science and Information Technology (SCQP09).

How the 225 questions were classified

1. **Page rendering and OCR.** Each page was rendered at 120 dpi; RapidOCR (ONNX Runtime) extracted the question and option text from the embedded images.
2. **Question segmentation.** PDF text-layer anchors (“Question Number”) and OCR geometry were combined to split each paper into 75 questions.
3. **Manual classification.** Every question was read and assigned to one subject → topic → subtopic. The classification file is kept alongside this document.
4. **Aggregation.** Counts were computed directly from the classification, not estimated. The three-year totals sum to 225.

Known limitations

- OCR is reliable for question stems but can garble mathematical notation and Hindi transliteration. Classification was done by reading the question, not by keyword matching alone.
- Some questions are cross-topic (for example, a DSA question using OS scheduling). They were assigned to the dominant concept.
- Format percentages are approximate (keyword tagging plus manual review), and weightage is a three-paper average, not a guarantee.
- The 2026 general-awareness questions are treated as an anomaly because they are absent in 2024 and 2025.

How to use this document with the question bank

The companion file **01_question_bank_recent.csv** contains all 225 questions with year, question number, subject, topic and subtopic. Use it to drill one module at a time: filter by subtopic, attempt the questions, then return to the card and check whether you can explain every bullet under “What to learn”.

The one-sentence summary

Master the top 20 modules, keep TDM on a daily drip, train the match/arrange/assertion formats, and protect your time from CN/DBMS and general awareness. That is the whole paper.

Prepared 1 Nov 2026 · Independent analysis of public NTA papers · Not affiliated with NTA · Verify the current syllabus.